Why do adult trees do not grow more with longer growing seasons and juveniles do?

by

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The following individuals certify that they have read, and recommend to the Faculty of Graduate and Postdoctoral Studies for acceptance, the thesis entitled:

Why do adult trees do not grow more with longer growing seasons and juveniles do?

submitted by **Christophe Rouleau-Desrochers** in partial fulfillment of the requirements for the degree of **Master of Science in Forestry** in **Department of Forest & Conservation Sciences**.

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Abstract

1. Shifted phenology with ACC extends GS; expected growth increase
2. Recent studies failed to find this relationship; with drought and competition restraining growth
3. Objective: understand if trees grow more with longer GS where drought and competition were limited
4. Leveraging 9 years of phenology from 2 studies, both in an arboretum and by relating it to tree growth
5. Juveniles grew more; adults did not change their growth with longer GS
6. Our results reveal a more complex growth response to climate change than currently assumed
7. Not massive responses and when they do, it's more for juveniles than mature: good for forest regeneration, potentially bad for forest carbon storage.

Lay Summary

Climate change extends the time during which plants can grow which is mainly caused by earlier springs and later autumns. For trees, a warmer and longer growing season means they should grow more. However, recent research showed that this may not be the case. Here we ask if juvenile and mature trees grow more with longer and warmer growing seasons. We will address this question using both young and adult broaleaf trees (Figure 1). Our results show that while young trees grew more and benefited from warmer and longer seasons, adult trees did not. This has important implication because juvenile trees are really important for forest regeneration, but they hold an insignificant proportion of the carbon stored in forests compared to adult trees. Our results thus partially support the recent observed decoupling between season length and growth which could have important implications for future carbon sequestration by forests.

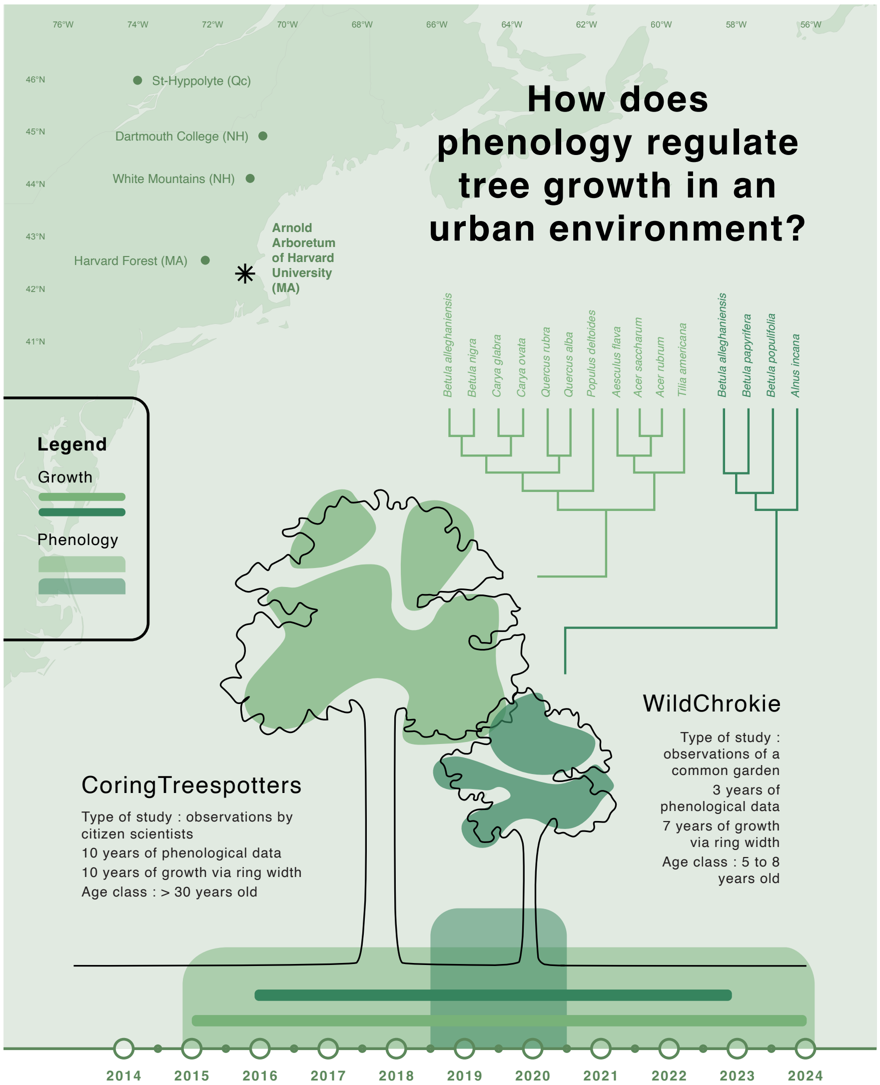

Figure 1: Chapter design including some nice stuff

Preface

1. Chapter 1
    (a) I leveraged phenology data collected by Dan Buonioto and Cat Chamberlain in Boston, a project started by Dr. EM Wolkovich
    (b) The research questions and project werer proposed by EMW
    (c) I processed, scanned, measured the tree cross-section and cores previously collected by Neil Pederson, Dan, EM Wolkovich and Deirdre Loughnan
    (d) Victor VdM and EM Wolkovich helped me with building the model
    (e) I performed the analyses and created all the associated figures
    (f) The following co-authors reviewed the manuscript: DL, VVDM, DB, CH, NP and EM Wolkovich.
    (g) No generative AI was used to write this thesis. I used GenAi for performing simple repetitive coding tasks and for help with the figures, data cleaning and improve coding workflow efficiency.
    (h) Should be submitted for publication by the time this gets sent to GPS
2. Chapter 2
    (a) I levereraged data collected from citizen scientists and cored the trees with XW
    (b) Project was conceptualized by EMW and me
    (c) I collected, processed, scanned and measured all the cores and performed the analyses
    (d) The following co-authors reviewed the manuscript: DL, VVDM, DB, CH, NP and EM Wolkovich.
    (e) GEN AI for this manuscript same as Chapt 1
    (f) This work is intended for publication

Table of Contents

List of Tables

List of Figures

Glossary

This glossary uses the handy acroynym package to automatically maintain the glossary. It uses the package's `printonlyused` option to include only those acronyms explicitly referenced in the LaTeX source. To change how the acronyms are rendered, change the `\acsfont` definition in `diss.tex`.

Acknowledgments

Thank those people who helped you.

Don't forget your parents or loved ones.

You may wish to acknowledge your funding sources.

Chapter 1

Warmer and longer growing seasons increase growth for juvenile trees in a common garden

Introduction

The most frequently observed biological impact of climate change is shifts in phenology—the timing of recurring life history events (Calvin *et al.*, 2023; Menzel *et al.*, 2006; Parmesan *et al.*, 1999). Across many plant species, spring events have advanced (Chmielewski & Rötzer, 2001; Fu *et al.*, 2014; Wolfe *et al.*, 2005), while autumn events have delayed—but to a much lesser extent (Gallinat *et al.*, 2015; Jeong & Medvigy, 2014). Together, early springs and later autumns lengthen the growing season (GS), which should increase growth for many species, including trees (Keenan *et al.*, 2014; Stridbeck *et al.*, 2022).

How much trees increase growth with longer seasons is important given the major role of forests in carbon sequestration (Friedlingstein *et al.*, 2026). Temperate forests account for roughly 40% of the annual global carbon sink, diminishing the consequences of human-emitted greenhouse gas emissions on the climate (Gibbs *et al.*, 2025). With warming, research has repeatedly found a connection between longer seasons and increased carbon storage (Boisvenue & Running, 2006; Finzi *et al.*, 2020; Gao *et al.*, 2022). The future of this net carbon sink by temperate forests, however, relies on the assumption that longer and warmer seasons will keep increasing carbon storage by trees (Pan *et al.*, 2024). A shift in this relationship could impact the future of this critical carbon sink.

New studies suggest that longer GS do not always increase growth (Cabon *et al.*, 2022; Charlet De Sauvage *et al.*, 2022; Dow *et al.*, 2022; Matula *et al.*, 2023; Silvestro *et al.*, 2023). Recently, Dow *et al.* (2022) found that warmer springs lengthened the growing season, but did not increase the peak growth rate or total annual growth in two temperate forest communities. Working across biomes, Cabon *et al.* (2022) found a similar trend, where gross primary production and growth appeared decoupled. However, neither of these studies could clearly establish why they found no relationship while previous studies have. Understanding whether temperate forests will persist as a net carbon sink requires answering why trees do not grow more despite longer GS.

A longer GS does not necessarily mean more favorable thermal conditions for growth. While the well-documented advance of spring phenology elongates the GS (Keenan *et al.*, 2014), spring days are short and usually cool (Figure 1.1a). Therefore, this advance may contribute little to a tree's annual growing degree days (GDD) accumulation (Figure 1.1b). In contrast, even small shifts to later end of season events can lead to larger changes in annual GDD; since more GDD accumulates per day during this period, any extra end of season (EOS) days may be more important than additional start of season days (SOS; Buonaiuto *et al.*, 2026).

Other factors beyond the thermal conditions may affect whether longer seasons increase growth. While trees may have a longer period to grow, water availability may become insufficient to support continuous growth throughout the GS (Wang *et al.*, 2025). Competition for light and nutrients may further limit the capacity of trees to benefit from longer seasons, especially for the ones positioned below the forest canopy (Augspurger & Bartlett, 2003; Cleland & Wolkovich, 2024; Sweeney *et al.*, 2024). Additionally, trees may have internal constraints that prevent them from increasing growth with longer seasons. Growth, like other biological processes, is under strict genetic control. Thus, even with optimal environmental conditions, physiological constraints may prevent temperate trees from growing indefinitely (Körner *et al.*, 2023; Wolkovich *et al.*, 2025), but we currently have a limited understanding of how well tree phenology and growth can track changing conditions.

Current research suggests that leaf and wood phenology differ in their plasticity in response to changing environmental conditions across species and life stages (Dox *et al.*, 2022). Leaf phenology (e.g. leafout) in the spring is a highly plastic trait with substantial variation across years with different climate, and also across species. Within a single community leafout can vary across species by over a month, with a general trend of species with more conservative growth strategies leafing out later (Loughnan *et al.*, 2025). Additionally, leaf phenology and growth appear closely synchronized in diffuse-porous species (e.g., *Betula;* Čufar *et al.*, 2008) and less so in ring-porous species (e.g., *Oaks*; González-González *et al.*, 2013), making leaf phenology a good proxy for setting the start and end of the GS in diffuse-porous species (Marchand *et al.*, 2021; Silvestro *et al.*, 2025; Suzuki *et al.*, 1996). While this proxy is useful for setting the boundaries of the GS, it does not indicate how much trees grow annually. How flexible growth is in response to changing GS length and conditions varies between species and populations, and may also differ with tree age, with juvenile trees being generally more flexible and responsive than their adult forms (Augspurger & Bartlett, 2003; Way & Montgomery, 2015).

Local adaptation could limit tree growth responses of different populations with future climate

(Soolanayakanahally *et al.*, 2013; Wolkovich *et al.*, 2025). While leaf phenology (e.g. leafout) in the spring is highly plastic and driven by temperature, end of season phenology (e.g. budset) is much less plastic and mostly driven by photoperiod (Baumgarten *et al.*, 2026; Singh *et al.*, 2017), with evidence of local adaptation (Zeng & Wolkovich, 2024). Provenance trials have repeatedly found trees from northern latitudes set bud earlier, even when grown in warm climates with longer growing seasons (Aitken & Bemmels, 2016; Way & Montgomery, 2015). This local adaptation and related reliance on photoperiod to trigger budset (an EOS event) likely explains why shifts in EOS with warming have been so much smaller than shifts in SOS. This predicts a latitudinal gradient in growth responses, with more limited responses in poleward populations where EOS is constrained to happen earlier in the season. Thus, common gardens investigating the effect of provenance latitude could help predict how climate change will affect tree growth.

To understand the relationship between GS and growth, we need to understand how growth varies across species and populations while disentangling how much longer seasons are also more thermally favorable seasons. Here, we address this by studying how growth and GS relate across species, populations and multiple metrics of GS length. Working with juvenile trees sourced from four populations and grown in one common garden, we examine how growth, measured via tree rings, and leaf phenology covary over three years. We consider effects of both the length of the season (i.e. number of growing days in the season) and its thermal conditions (i.e. GDD accumulated in the season).

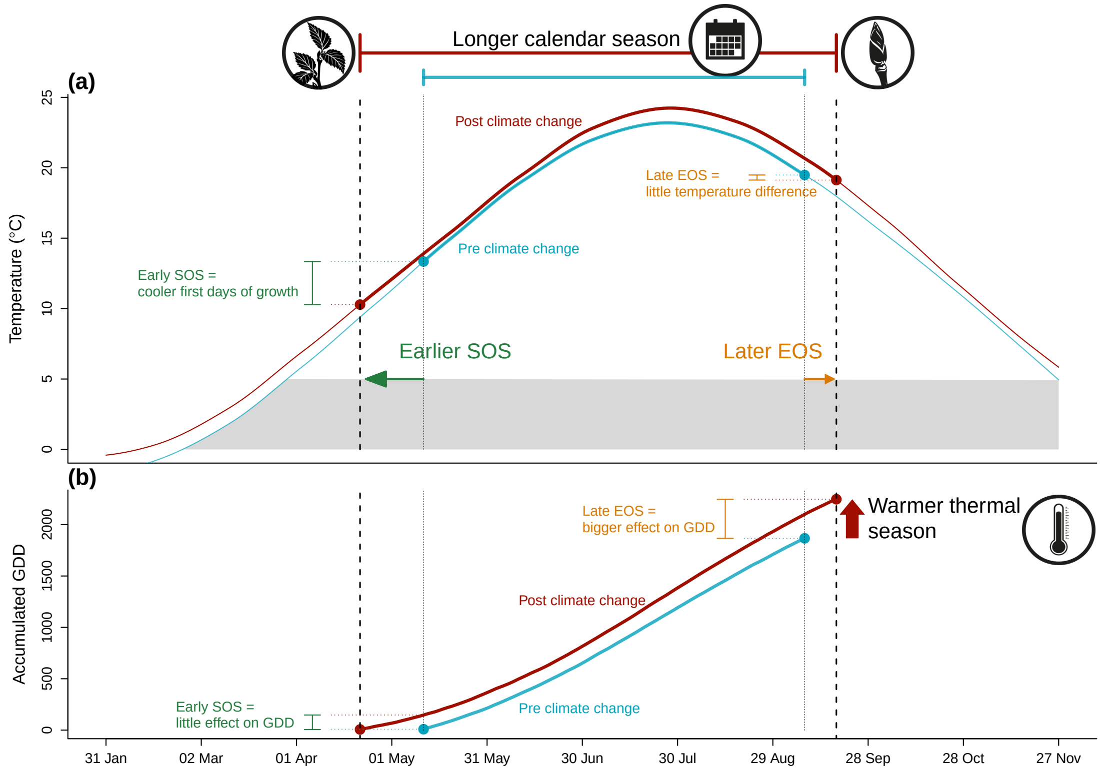

Figure 1.1: Shifting growing season lengths with climate change affect both the temperature trees experience throughout the growing season (a), and how many growing degree days (GDD) they accumulate (b). Here we show changes in daily (smoothed) and accumulated temperatures before (1945-1965, in blue) and significant anthropogenic climate warming (2005-2025, in red) using data local to our study. The arrows represent hypothetical shifts in spring (green) and fall (orange) phenology. With climate change, spring events have advanced and fall phenology has delayed, but to a much lesser extent—lengthening the calendar growing season. This concurrently changed the temperature trees experience early in their season, with cooler first days of growth (a) causing little accumulation of GDD (b). In contrast, weaker shifts in EOS and little temperature change (a) results in a relatively high GDD accumulation (b) due to warmer temperature at this time of the season. Alongside these SOS and EOS shifts, warmer summer temperatures increased GDD accumulation, further warming their thermal seasons. We used mean daily temperature data from the Boston Logan International Airport Climate station for the smoothed temperature and GDD curves (National Oceanic and Atmospheric Administration (NOAA), 2026).

Methods

Common garden setup and data collection

We collected seeds from 18 species of deciduous trees and shrubs from four provenances across a 3.5° latitudinal gradient (Figure S1; Table S1) in 2014-2015, germinated them following standard germination protocols and grew them to seedlings (Baskin & Baskin, 2014). In the spring of 2017, we planted the seedlings outside at the Weld Hill Research Building at the Arnold Arboretum in Boston MA (42.30°N, 71.13°W). Plots were regularly weeded and watered throughout the duration of the study and were pruned in the fall of 2020 (Buonaiuto *et al.*, 2026).

For the years 2018-2020, we monitored leaf phenology for all individuals in the common garden once or twice per week from February to December; the exact frequency varied within and across years (for more details, see Figure S1). For this, we used a modified BBCH scale to observe two phenological stages: leafout which we defined as: BBCH 15 and budset as BBCH 102; (Buonaiuto *et al.*, 2026; Finn *et al.*, 2007). We had a total of 239 leafout and 230 budset observations (which we refer to as the 'full datasets'), but for some individual trees in certain years we did not have both events, thus our total of matched full growing season duration (which we refer to as the 'restricted dataset') is 227 observations.

In the spring of 2023, we collected tree ring samples from tree species with high survivorship. Thus, we have four species in the *Betulaceae* family : Grey alder (*Alnus incana*), Yellow birch (*Betula alleghaniensis*), Paper birch (*Betula papyrifera*) and Grey birch (*Betula populifolia*). We collected cross-sections on 78 individuals by cutting the trunk above the root collar, and cores on 43 individuals using standard dendrochonological methods (Cook & Kairiukstis, 1990), for a total of 81 individual trees (Table S1). We stored the cores in modified plastic straws to promote aeration and left them to dry with the cross-sections at ambient temperature for three months.

Sample processing, imaging and measuring

We mounted cores on wooden mounts, and sanded both the cores and cross-sections using progressively finer sandpaper grits (150, 300, 400, 600, 800, 1000), then scanned the cores and cross-sections at a resolution of 6250 dpi with a high-resolution tree ring scanner (Fong *et al.*, 2026). Using the digitized images, we measured ring widths with ImageJ (Schneider *et al.*, 2012) at three standard locations for each cross-section (at 120° from each other, averaging the three measurements to yield one ring width per year, per individual), and one ring width for each core. When we had cross-sections ($n = 78$) and cores ($n = 43$) for an individual, we used only the measurement from the cross sections ($n = 81$).

We performed visual cross dating. This consists of assigning a calendar year to each ring and comparing the year-to-year growth variation and anomalies across replicates of the same species (Cook & Kairiukstis, 1990). We did not perform statistical crossdating or detrending because our

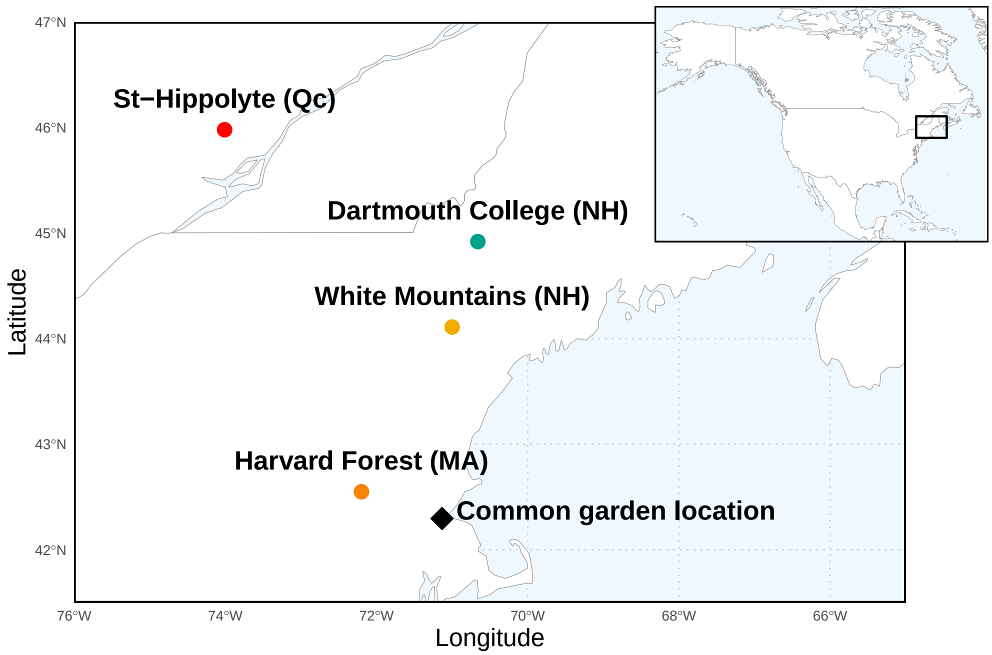

Figure 1.2: Provenance where the seeds were sourced from are depicted as circles, while the diamond denotes the location of the common garden. The common garden location is where the study took place at the Weld Research Building (Boston, MA). The colors of the dots match the model estimates of Figure 1.4.

chronologies (number of years of data) were extremely short (Raden *et al.*, 2020) and showed no evidence of consistent allometric trends (Figure S1). We controlled for effects of different years by including 'year' in our models (see equation 2.1 in 'Data analyses' below).

Data analyses

To understand how growth shifted with season length we used Bayesian hierarchical models, estimating how growth, measured as ring width, covaried with season length, measured in four different ways: GDD, GSL, SOS and EOS. We calculated the thermal season using growing degree days (GDD; mean daily temperature above 5°C) accumulated between leafout and budset, and the calendar season (or growing season length; GSL) as the number of days between leafout and budset. Finally, we used leafout as the start (SOS) and and budset end of the season (EOS) as separate predictors that represent the boundaries of the growing season. For interpretability, we report ring width change (log(mm)) as the number of GDDs accumulated in an average week (seven days) in May (day of year 121 to 150 from 2018 to 2020), for the thermal season, and ring width change per week of season length, leafout and budset for GSL, SOS and EOS, respectively. We report effects when the 90% UI do not overlap with zero and consider a trend if the 50% UI do not overlap with zero.

For daily climate data across all the years of phenological sampling (2018-2020) we combined two sources. For 2018 and 2019 we used data from the climate station located at Weld Hill research station (Arnold Arboretum, 2026). Because the data was not available for that station after August 2020, we used data from the weather station 'Boston (Weld Hill)' of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (Carroll *et al.*, 2026). We used the North American Drought Monitor Maps to determine drought occurrence across year at our study location and used the proposed drought intensity index to qualify the presence of moderate, severe or extreme droughts (National Centers for Environmental Information (NCEI), 2026).

For each predictor, we estimated its effect on observed ring width (i, denoting each observation) for each tree ($treeid$) in each year ($year$) as:

$$\begin{aligned} ln(ringwidth)_i &\sim Normal(\mu, \sigma_y) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{prov}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\ \alpha_{\text{prov}} &\sim Normal(0, \sigma_{\alpha\,\text{prov}}) \\ \alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha\,\text{treeid}}) \end{aligned} \tag{1.1}$$

This model estimates a species-specific effect of each season length predictor (e.g. GDD) on growth ($\beta_{species}$), while also controlling for other variation due to species ($\alpha_{species}$), provenance (α_{prov}), individual tree (α_{treeid}) and year (α_{year}). We ran the models using both natural units, and also using scaled predictors (via z-scoring. Gelman & Hill, 2006) to allow for direct comparison of effect sizes across different metrics. We report estimates as means and 90% uncertainty intervals (UI).

We used weakly informative priors for all models (increasing the priors by 2× for the z-scored models did not qualitatively affect the results). We ran four chains with 2000 warm-up and 2000 sampling iterations each, for a total of 8000 sampling iterations and checked diagnostics of output: chains had no divergent transitions, $\hat{R}$ values below 1.01, and effective sample sizes (n_{eff}) greater than 10% of the model iterations. We also fit the models for SOS and EOS using both the restricted (n = 227) and full datasets for each predictor (SOS n = 239 and EOS n = 230. We wrote our models using Stan (Carpenter *et al.*, 2017) with the rstan package version 2.32.7 (Stan Development Team, 2025) to run the Stan code in R.

Because ring widths decrease with age—particularly with young trees—we also ran the models using basal area increment (BAI) as the response variable. This allows for a more representative portrayal of annual growth increment by using the total ring area, instead of its width alone. To calculate BAI we summed each ring width separately at the three distinct locations on the cross sections yielding three radii from pit to bark. We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring: $\pi \times$ (inner radius^2 − outer radius^2), and averaged across the three BAI for each ring.

Results

Our dataset comprised four deciduous tree species from four sites with leaf phenology data spanning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals. At the common garden location, the mean summer temperature was similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C). There was no drought for those three years, except for in 2020, where a moderate summer drought (June-August) ramped up to a severe drought in September (National Centers for Environmental Information (NCEI), 2026). The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018; Figure S8a and b). The earliest leafout was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020; Figure S8c). The earliest budset was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018; Figure S8d. Finally, annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018; Figure S7).

Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it. We report estimates as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henceforth 'RW/day' or 'RW/GDD,' with days and GDD being adjusted to represent more meaningful units, as described above (see 'Data analyses' section in Methods for more details). Warmer thermal seasons increased growth for Paper birch (*Betula papyrifera*; $\beta = 0.15$, UI: 0.06, 0.23 RW/GDD), Grey birch (*Betula populifolia*; $\beta = 0.07$, UI: 0.02, 0.12 RW/GDD), and Yellow birch (*Betula alleghaniensis*; $\beta = 0.04$, UI: 0.00, 0.08 RW/GDD), grey alder (*Alnus incana*) trended in the same direction ($\beta = 0.02$, UI: -0.01, 0.05 RW/GDD); Figure 1.3a and e; TableS2). Unlike its response to warmer thermal season, *A. incana* grew more with longer calendar seasons ($\beta = 0.05$, UI: 0.00, 0.10 RW/day). *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.18$, UI: 0.05, 0.31 RW/day), while the other two *Betula* did not, though they trended in the same way (*B. alleghaniensis*: $\beta = 0.03$, UI: -0.03, 0.09 RW/day, and *B. populifolia*: $\beta = 0.05$, UI: -0.02, 0.13 RW/day; Figure 1.3b and f; Table S2). On a scaled axis (through z-scoring, Gelman & Hill, 2006), the thermal and calendar season were similarly predictive (Figure S11). Results with basal area increment (BAI) for the GDD model were similar to ring width (see Table S5).

Increased growth with longer seasons is often presumed to come from growth in the spring, since spring phenology has shifted much more than fall phenology. In contrast to this expectation, we found most species grew more with a later end of season, and only one species grew more with an earlier start of season (*A. incana* (one of the two species that a longer calendar season increased growth for: $\beta = -0.07$, UI: -0.13, -0.01 RW/day; Figure 1.3c and g; Table S2). *B. populifolia* trended in the opposite direction ($\beta = 0.05$, UI: -0.07, 0.16 RW/day; Figure 1.3c and g; Table S2). *B. alleghaniensis*, which did not grow more with longer calendar season, grew less with an earlier start of season ($\beta = 0.10$, UI: 0.00, 0.20 RW/day; Figure 1.3c and g; Table S2)—but more with a later end of season (β = 0.12, UI: 0.05, 0.18 RW/day). Alongside *B. alleghaniensis*, two other species grew more with a later end of season (*B. populifolia*: β = 0.10, UI: 0.02, 0.17 RW/day, and *B. papyrifera*: β = 0.21, UI: 0.09, 0.33 RW/day; Figure 1.3d and h; Table S2), with the latter being the only one of those three that also grew more with longer calendar season. The estimates were unchanged when using the full (SOS: n = 239 and EOS: n = 230) or restricted datasets (n = 227; Figure S3a).

We did not find any clear effect of year on growth (2018: α = 0.19, UI: -0.77, 1.15 RW; 2019: α = 0.18, UI: -0.79, 1.14 RW; 2020: α = -0.43, UI: -1.40, 0.53 RW; Table S3), and there were no apparent differences in baseline growth between species (*A. incana*: α = 1.06, UI: -3.19, 5.21 RW; *B. alleghaniensis*: α = 0.21, UI: -3.97, 4.34 RW; *B. papyrifera*: α = -4.00, UI: -8.89, 0.75 RW; *B. populifolia*: α = -0.66, UI: -4.94, 3.63 RW; Figure S4 and S5; Table S3). In addition, growth did not vary with provenance (Dartmouth College (NH): α = -0.03, UI: -0.22, 0.14 RW; Harvard Forest (MA): α = -0.10, UI: -0.33, 0.07 RW; St-Hippolyte (Qc): α = 0.04, UI: -0.14, 0.24 RW; White Mountains (NH): α = 0.08, UI: -0.09, 0.29 RW; Figure 1.4; Table S3). The limited variation between provenances (σ = 0.17, UI: 0.02, 0.48 RW) was similar to variation across individual trees (σ = 0.17, UI: 0.05, 0.27 RW; Table S3).

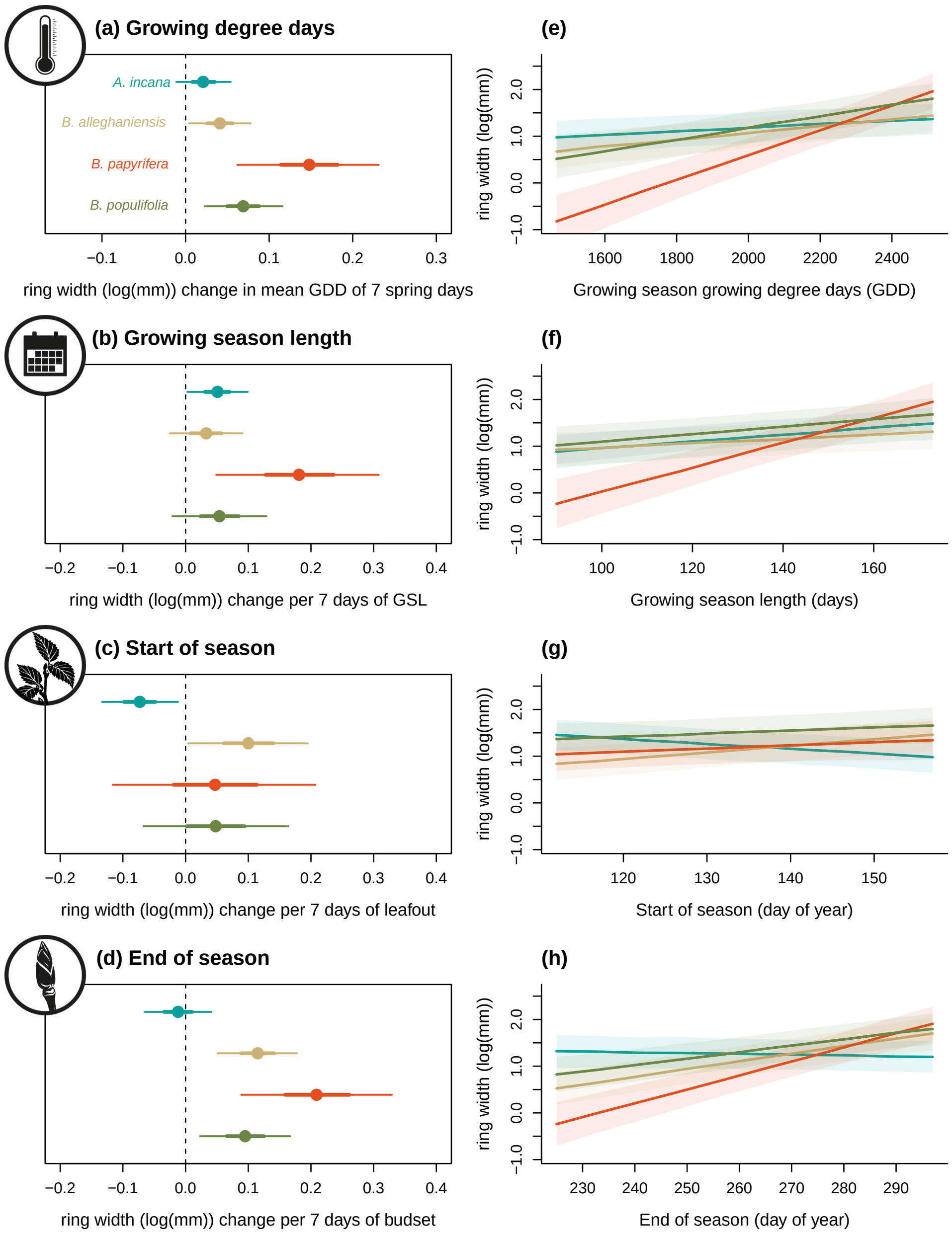

Figure 1.3: Estimates of ring width change (log(mm)) as a function of four different phenological predictors: growing degree days of the growing season (mean daily temperature above 5°C); growing season length, start of season, and end of season across four species: Grey alder (*Alnus incana*), Yellow birch (*Betula alleghaniensis*), Paper birch (*Betula papyrifera*) and Grey birch (*Betula populifolia*; species colors are the same across the panels). Panels on the left (a to d) represent the mean model slope estimates for each species (dots) with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively), for each predictor, which we represent through ring width change in mm on the log scale per adjusted predictor (see 'Data analyses' section in Methods for more details on predictor adjustment). See Table S2 for parameter estimates. Panels on the right (e to h) show the estimated response of ring width as a function of each predictor, with the lines depicting the mean response at each predictor value, and the shaded area, the 50% uncertainty intervals.

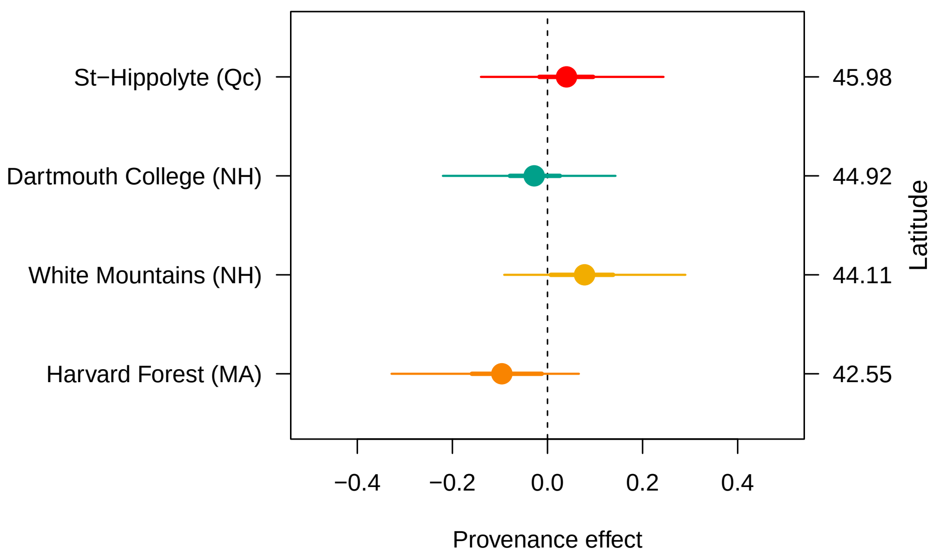

Figure 1.4: The effect of provenance on growth across a latitudinal gradient. We show the ring width (log(mm); see Table S3 for all parameter estimates) for each provenance with growing degree days as the predictor (effects were similar for the other predictors, see Table S3). The dots represent the estimated mean of each provenance, with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively). The provenances are color-coded from Figure S1, the map showing their locations.

Discussion

Here we used 81 juvenile trees of four species from four provenances to study how longer growing seasons affect tree growth (using annual growth via tree rings and leaf phenology as a proxy for the start and end of season, SOS and EOS, respectively). Longer GS increased growth for all species, but varied depending on the season metric used to measure it, with the thermal growing season more often relating to growth than the calendar season.

Our results support the idea that the thermal season is be a better predictor for growth (Chuine & Régnière, 2017; Wang & Engel, 1998). By accounting for differences in daily temperatures, the thermal season may be more closely aligned to when and how fast plants can grow each day compared to the calendar season, which assumes a stable influence of each day on growth. Thus, the short and cool spring days early in the season count for less in the thermal season than the calendar season (Buonaiuto *et al.*, 2026). Further supporting this, when we disentangled the calendar season by using the SOS and EOS separately, we found a larger response of growth to EOS, with little effect of SOS.

The finding that EOS appears more related to growth than SOS is aligned with increasing work suggesting SOS and EOS together may dynamically determine growth (Zohner *et al.*, 2023). This implies that research focusing solely on SOS (Dow *et al.*, 2022; Gao *et al.*, 2022; Wheeler *et al.*, 2016) could miss when longer seasons lead to more growth, and also that the strong focus on SOS in phenology research may ignore an important part of the season for growth. While studies have long called for more research on EOS (Gallinat *et al.*, 2015; Gunderson *et al.*, 2012; Piao *et al.*, 2019; Richardson *et al.*, 2012), our results suggest we may especially need more work that monitors how growth and phenology shift across the season (Zani *et al.*, 2020; Zohner *et al.*, 2023). More studies using data with a finer temporal scale (e.g., daily or weekly data through dendrometers and/or xylogenesis) could confirm if most growth occurs closer to EOS versus SOS. To date, these studies showed strong species differences as to when most of the growth happens in the season. However, most of these studies were done on a small number of European and/or conifer species (e.g. Balducci *et al.*, 2019; Etzold *et al.*, 2022; Zweifel *et al.*, 2016). More data across species spanning a wider range of life history strategies could elucidate how dynamically growth can change throughout the season.

Our results suggest different growth strategies between species may yield different responses to longer seasons. Our most ruderal species, *A. incana* was the only species to grow more with a longer calendar season and an earlier SOS. This suggests that species with more ruderal, flexible life-history strategies may actually take advantage of growing more in the spring despite cooler, shorter days (Bowman & Hacker, 2023; Furlow, 1997; Grime, 1977). In contrast, *B. papyrifera* and *B. populifolia* which are fast growing, but more conservative than *A. incana*, were both most driven by the EOS, suggesting that they would be able to exploit these late season days characterized by warmer temperatures (Burns *et al.*, 1990). Lastly our slowest growing, longest living species, *B. al-*

leghaniensis, responded to both the SOS and the EOS. This points towards a possible 'neutralizing' response where an early SOS decreases growth, but a later EOS increases it, muting this species' response to a longer calendar season. Studies including an even wider span of growth strategies across species may find further differences. Our four study species are all relatively fast growing, which should allow them to track changing conditions better than their slower growing counterparts. While we found an overall positive trend of warmer seasons on growth, this might have changed if we used slower growing, more conservative species (such as *Quercus*; Burns *et al.*, 1990). Additionally, we used juvenile trees which tend to be more reactive and could in part explain why they grew more with warmer seasons (Augspurger & Bartlett, 2003; Sweeney *et al.*, 2024). While other studies on mature trees have found the same result, our results may have been more muted if the trees were grown in a forest setting. We grew trees in a common garden setting with potentially more space between trees (spacing was between 1.5 and 3 m), without any shade cloth. If these trees grew in a mature forest they would have been below the canopy, competing for light and other resources and thus might not have responded as much as we observed in our competition-limited conditions (Polgar & Primack, 2011). However, our results may have implications beyond how mature forests function.

Our focus on juvenile trees may suggest how forests regenerate and assemble with the longer warmer seasons of anthropogenic climate change. Plant phenology research has suggested earlier seasons with warming favor the species that shift earlier the most, with many studies focused on invasive species appearing to support this (Wolkovich *et al.*, 2013; Zettlemoyer *et al.*, 2019). Yet, these studies are often on herbaceous species and our results, along with other recent work (Zani *et al.*, 2020; Zohner *et al.*, 2023), suggest mid- and later seasons effects may have under-appreciated performance implications in woody species. Thus, more studies of how later season events correlate with invasions, and forest assembly more generally, could be especially important for understanding how warming will shift forest regeneration.

Our results also suggest assembly models may need less of a focus on population-level differences than previously suggested (Cavender-Bares, 2019). While studies of large latitudinal gradients have found lower growth in tree species from poleward latitudes (Aitken & Bemmels, 2016; Soolanayakanahally *et al.*, 2013), we found no differences in growth across a latitudinal gradient of 3.5°. Thus, at least across this latitudinal range, local adaption on growth appears weak. Supporting this, we found no evidence of shifting EOS across provenances, despite decades of work finding that EOS events, such as budset, are usually more locally adapted than SOS events such as leafout (Soolanayakanahally *et al.*, 2013; Zeng & Wolkovich, 2024). Yet, a study of the same common garden including all tree and shrub species found EOS appeared to vary strongly by year and shift earlier with earlier SOS (Buonaiuto *et al.*, 2026), suggesting EOS may be more variable than currently proposed (Alberto *et al.*, 2013). Taken together these results imply that our current understanding of budset being strongly locally adapted and thus varying little with temperature or SOS needs more study.

Implications for large-scale vegetation modeling

Using a rarely available dataset of tree growth coupled with leaf phenology SOS and EOS, we found an overall positive response from multiple tree species to longer seasons, which has important implications for models of future carbon storage. While juvenile trees store little carbon compared to their adult forms, they are crucial for forest regeneration capacity and range expansion (as discussed above, and see Luu *et al.*, 2024), and generally predict adult tree responses qualitatively (Wolkovich *et al.*, 2025). We found that SOS and EOS asymmetrically matter to growth, which means that how we model these phenological events in vegetation and related earth system models may be critical for accurate predictions. Modelling EOS better will require more studies, especially on more species across a larger gradient. This could help tease out whether local adaptation of EOS depends on the species, latitudinal range (both distance and how far north provenances come from), or some combination of the two, which could then feed into improved models.

Integrating our results into models will also require considering how environmental conditions beyond temperature shift across the season, especially soil moisture (Gao *et al.*, 2022; Li *et al.*, 2023). Trees in our common garden were regularly watered, likely relieving any major drought pressures though studies beyond ours have found a connection between later EOS and more growth, (see: Gao *et al.*, 2022; Wang *et al.*, 2025). For the four species we studied, there are currently no increases in drought occurrence within their ranges (Agel *et al.*, 2025; National Centers for Environmental Information (NCEI), 2026), suggesting that their growth may increase with longer and warmer GS (Agel *et al.*, 2025; Cook & Pederson, 2011), as we found. However, the positive effect of a later EOS could be offset if soil moisture becomes inadequate for growth, and other temperate regions with decreased soil moisture have reported declines in tree growth (e.g., Chiang *et al.*, 2021; Spinoni *et al.*, 2018). Improved accuracy of future carbon sequestration models would consequently require integrating over more regions and growth drivers.

Supplemental material

Supplemental Methods

Table S1: Provenance coordinates and count per species and provenances out of a total of 81 individual trees sampled from the United States of America (in Massachusetts (MA) and New Hampshire(NH)) and Canada (in Quebec (Qc)).

Provenance	Longitude	Latitude	*A. incana* ($n = 29$)	*B. alleghaniensis* ($n = 20$)	*B. papyrifera* ($n = 7$)	*B. populifolia* ($n = 25$)
Harvard Forest (MA)	-72.2	42.5	7	0	4	7
White Mountains (NH)	-71.0	44.1	6	7	0	9
Dartmouth College (NH)	-70.7	44.9	8	7	2	9
St-Hippolyte (Qc)	-74.0	46.0	8	6	1	0

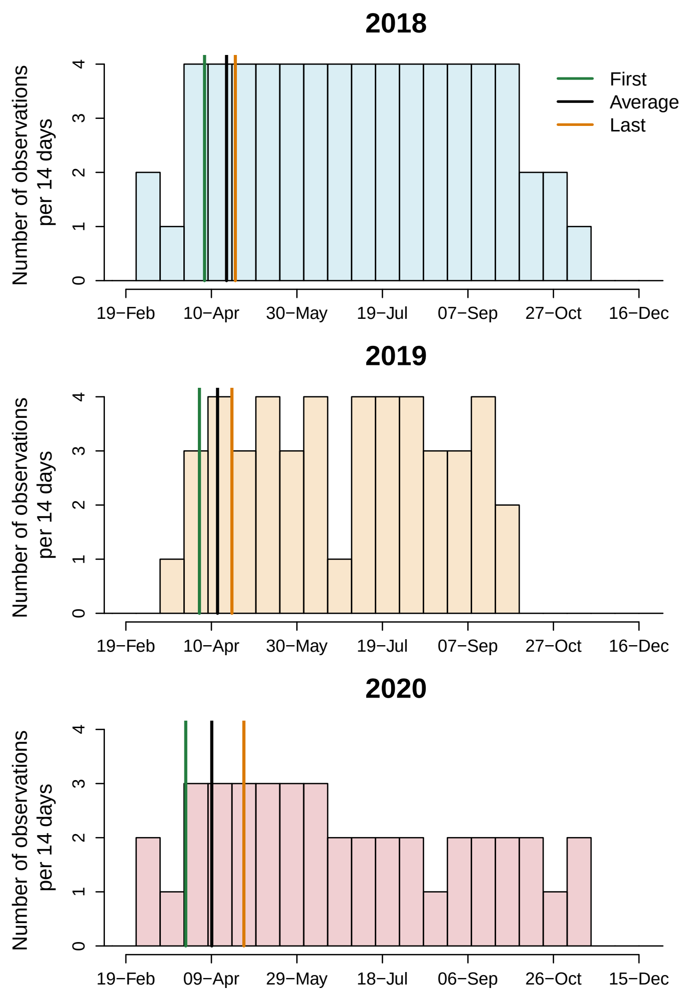

Figure S1: Leaf phenology monitoring frequency across years. We monitored leaf phenology once or twice per week for three years, though the frequency varied within and across years. Each histogram bin represents the number of observations that were made during a period of 14 days and the lines represent the earliest, the average and last recorded budburst observations. This demonstrates that across the different frequencies of observation, distances between early spring phenophases were similar across years.

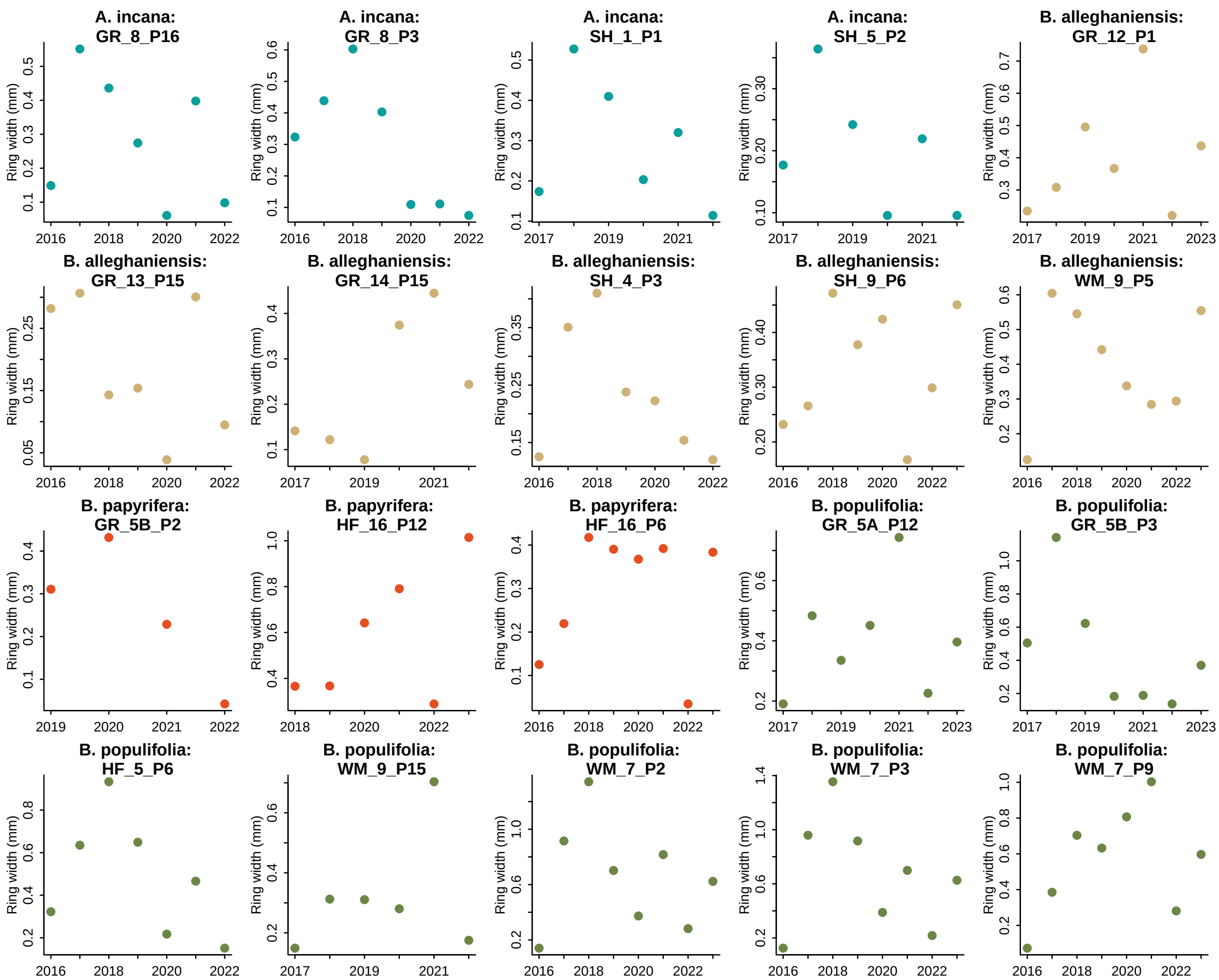

Figure S2: We show the potential allometry patterns across all years we had ring width for. For this, we randomly sampled 20 trees from our dataset and represent their ring width (mm) for each year.

Supplemental Results

Table S2: Species level slope parameters which represent ring width change (log(mm)), per adjusted predictor (see 'Data analyses' in Methods for more details). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different phenological predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

Species	GDD	GSL	SOS	EOS
A. incana	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.13, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.15 [0.06, 0.23]	0.18 [0.05, 0.31]	0.05 [-0.12, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.07 [0.02, 0.12]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.02, 0.17]

Table S3: The effect of provenance on ring widths (log(mm)). We report these intercept estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

Provenance	GDD	GSL	SOS	EOS
Harvard Forest (MA)	-0.10 [-0.33, 0.07]	-0.09 [-0.34, 0.09]	-0.04 [-0.26, 0.12]	-0.10 [-0.33, 0.07]
White Mountains (NH)	0.08 [-0.09, 0.29]	0.08 [-0.09, 0.31]	0.08 [-0.08, 0.29]	0.07 [-0.09, 0.27]
Dartmouth College (NH)	-0.03 [-0.22, 0.14]	-0.04 [-0.25, 0.15]	-0.06 [-0.27, 0.10]	-0.04 [-0.24, 0.12]
St-Hippolyte (Qc)	0.04 [-0.14, 0.24]	0.04 [-0.16, 0.26]	0.02 [-0.15, 0.22]	0.04 [-0.14, 0.24]

Table S4: Summary output of each parameter from our four predictors, using ring width (log(mm)) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See 'Data analyses' in Methods section for more details.

Parameter	GDD	GSL	SOS	EOS
α	-0.62 [-4.69, 3.53]	0.12 [-3.92, 4.13]	1.15 [-2.58, 4.83]	-1.29 [-5.36, 2.76]
$\sigma_{\alpha_{treeid}}$	0.17 [0.05, 0.27]	0.18 [0.06, 0.28]	0.21 [0.10, 0.30]	0.18 [0.06, 0.28]
$\sigma_{\alpha_{prov}}$	0.17 [0.02, 0.47]	0.18 [0.02, 0.48]	0.17 [0.02, 0.46]	0.17 [0.02, 0.46]
σ_y	0.48 [0.43, 0.53]	0.48 [0.44, 0.53]	0.47 [0.43, 0.52]	0.47 [0.42, 0.52]
$\beta_{species_{A.incana}}$	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.13, -0.01]	-0.01 [-0.07, 0.04]
$\beta_{species_{B.alleghaniensis}}$	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
$\beta_{species_{B.papyrifera}}$	0.15 [0.06, 0.23]	0.18 [0.05, 0.31]	0.05 [-0.12, 0.21]	0.21 [0.09, 0.33]
$\beta_{species_{B.populifolia}}$	0.07 [0.02, 0.12]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.02, 0.17]
$\alpha_{prov_{DartmouthCollege(NH)}}$	-0.03 [-0.22, 0.14]	-0.04 [-0.25, 0.15]	-0.06 [-0.27, 0.10]	-0.04 [-0.24, 0.12]
$\alpha_{prov_{HarvardForest(MA)}}$	-0.10 [-0.33, 0.07]	-0.09 [-0.34, 0.09]	-0.04 [-0.26, 0.12]	-0.10 [-0.33, 0.07]
$\alpha_{prov_{St-Hippolyte(Qc)}}$	0.04 [-0.14, 0.24]	0.04 [-0.16, 0.26]	0.02 [-0.15, 0.22]	0.04 [-0.14, 0.24]
$\alpha_{prov_{WhiteMountains(NH)}}$	0.08 [-0.09, 0.29]	0.08 [-0.09, 0.31]	0.08 [-0.08, 0.29]	0.07 [-0.09, 0.27]
$\alpha_{year_{2018}}$	0.19 [-0.77, 1.15]	0.24 [-0.70, 1.17]	0.20 [-0.75, 1.14]	0.22 [-0.71, 1.16]
$\alpha_{year_{2019}}$	0.18 [-0.79, 1.14]	0.10 [-0.83, 1.06]	0.11 [-0.84, 1.04]	0.11 [-0.83, 1.05]
$\alpha_{year_{2020}}$	-0.43 [-1.40, 0.53]	-0.34 [-1.27, 0.60]	-0.38 [-1.32, 0.56]	-0.49 [-1.42, 0.46]
$\alpha_{species_{A.incana}}$	1.06 [-3.19, 5.21]	0.11 [-3.92, 4.21]	1.48 [-2.13, 5.13]	3.06 [-1.40, 7.36]
$\alpha_{species_{B.alleghaniensis}}$	0.21 [-3.97, 4.34]	0.35 [-3.69, 4.53]	-1.91 [-5.64, 1.79]	-1.85 [-6.39, 2.59]
$\alpha_{species_{B.papyrifera}}$	-4.00 [-8.89, 0.75]	-2.61 [-7.17, 1.95]	-0.79 [-5.03, 3.36]	-5.57 [-11.02, -0.07]
$\alpha_{species_{B.populifolia}}$	-0.66 [-4.94, 3.63]	0.22 [-3.97, 4.45]	-0.53 [-4.29, 3.22]	-0.88 [-5.38, 3.67]

Table S5: Summary output of each parameter from our four predictors, using basal area increment ($\log(mm^2)$) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See data analyses section for more details.

Parameter	Estimate
α	0.62 [-4.52, 5.84]
$\sigma_{\alpha_{treeid}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{prov}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{species_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{species_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{species_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{species_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{species_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{species_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{species_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{species_{B.populifolia}}$	-0.07 [-5.98, 5.57]

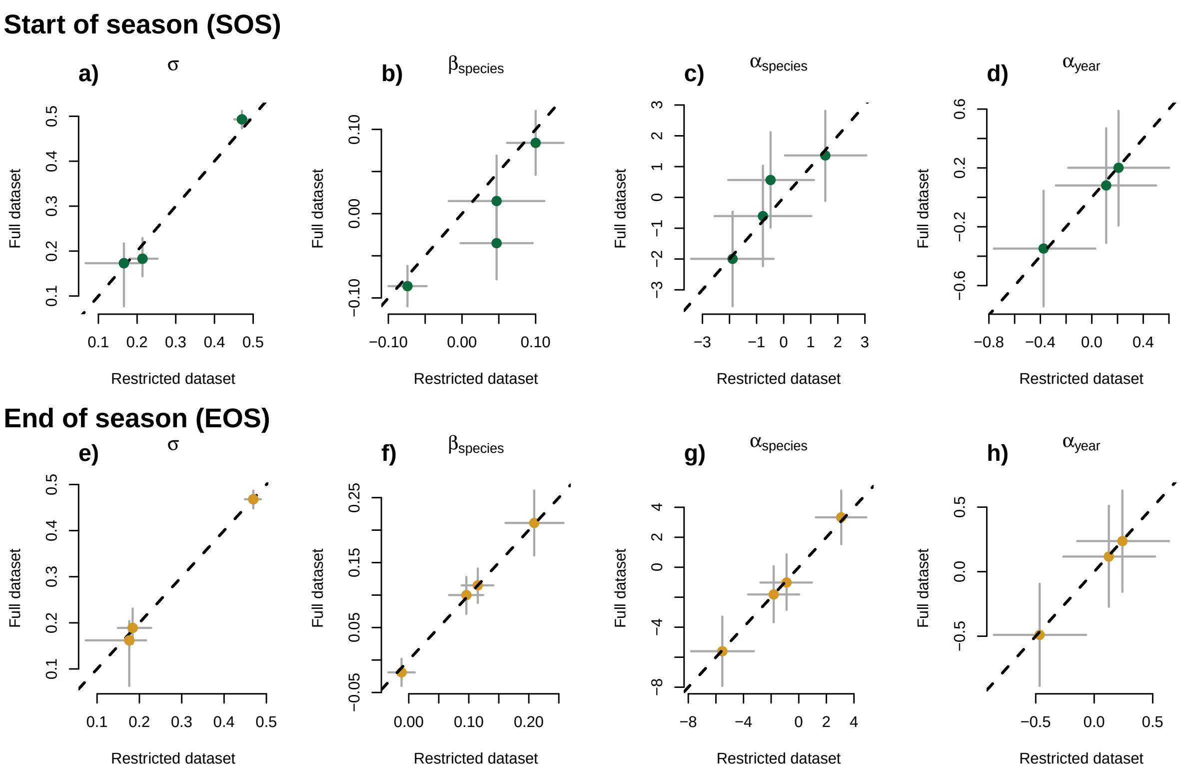

Figure S3: Full vs most restricted datasets for the model with start of season (a-d) and end of season (d-h) as the predictor. Each panel represent the different parameters used to fit the models. σ (a, e), $\alpha_{species}$ (c, g) and α_{year} (d,h) represent the ring width (log(mm)) for species and year, respectively. $\beta_{species}$ (b, f) represent the change in ring width (log(mm))/adjusted predictor, for each species (see 'Data analyses' section in Methods for more details).

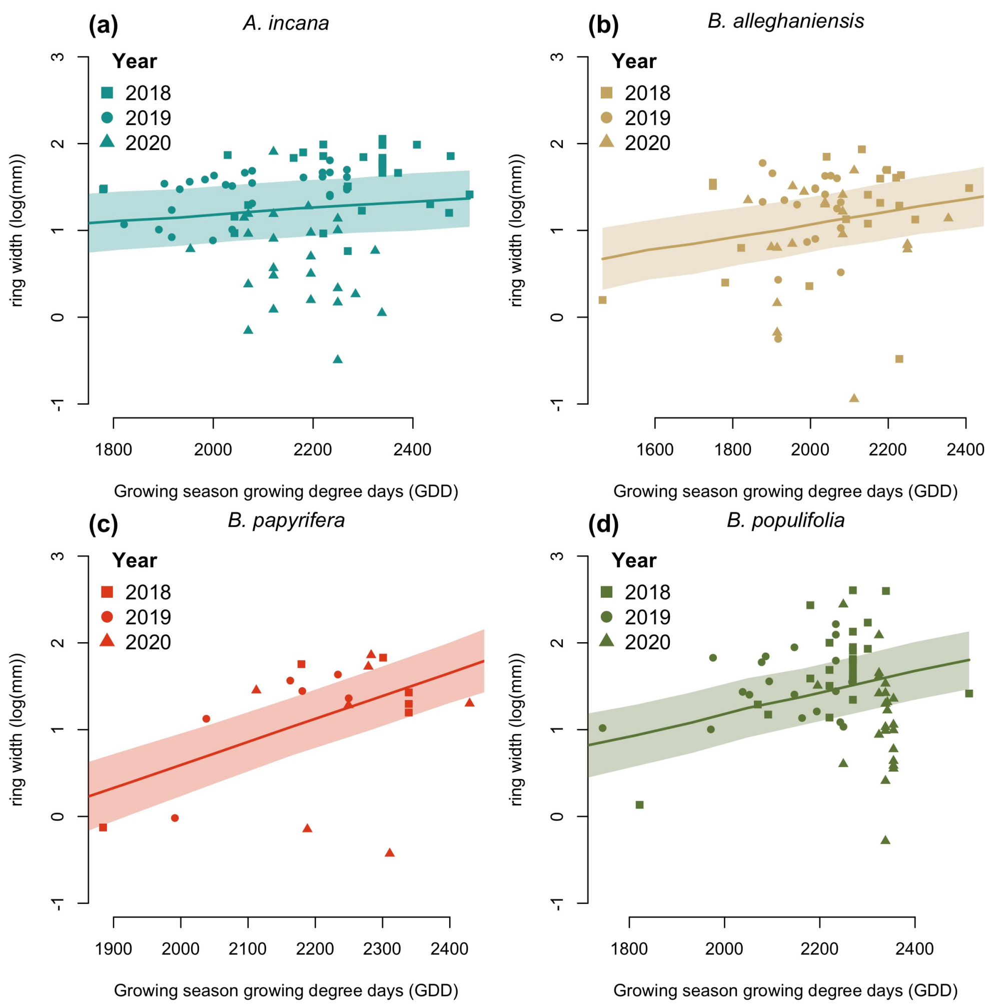

Figure S4: Ring width (log(mm)) trends in response to thermal season (growing degree days). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI at each predictor value. The dots are the empirical data shaped by year.

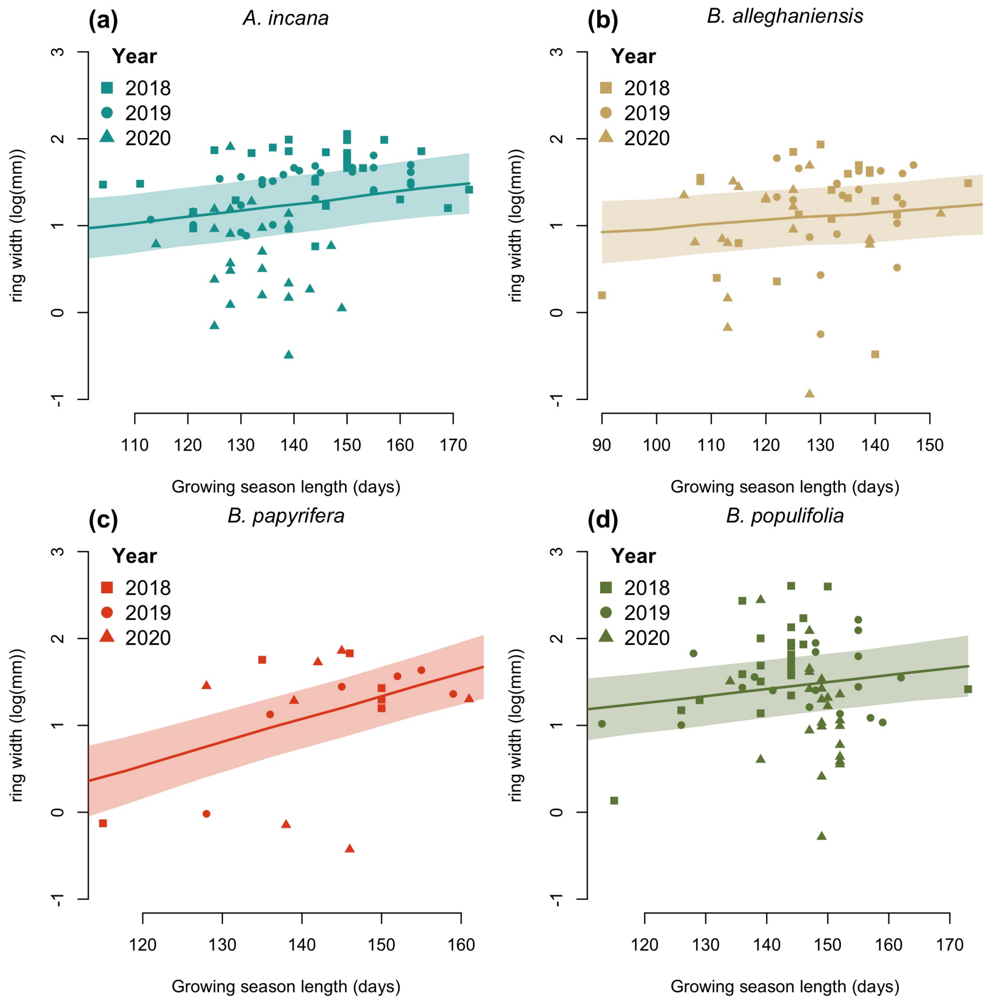

Figure S5: Ring width trends (log(mm)) in response to calendar season (growing season length). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI at each predictor value. The dots are the empirical data shaped by year.

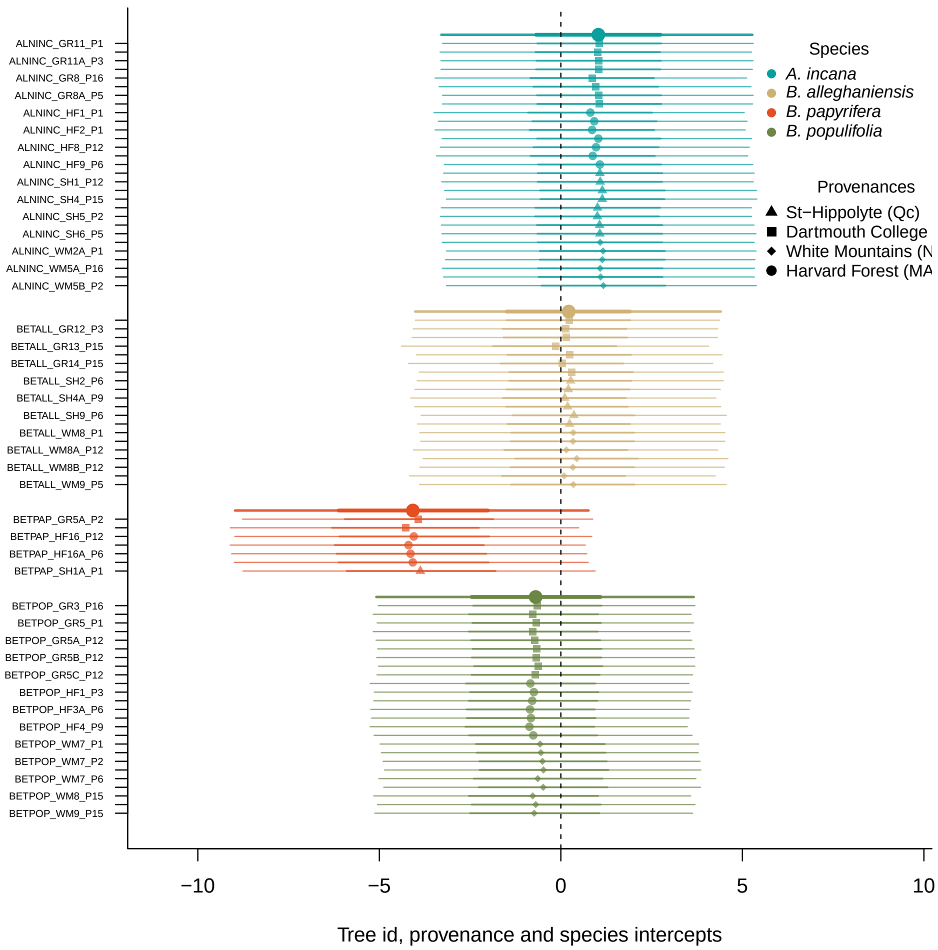

Figure S6: Ring width (log(mm)) deviations from the grand mean for grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

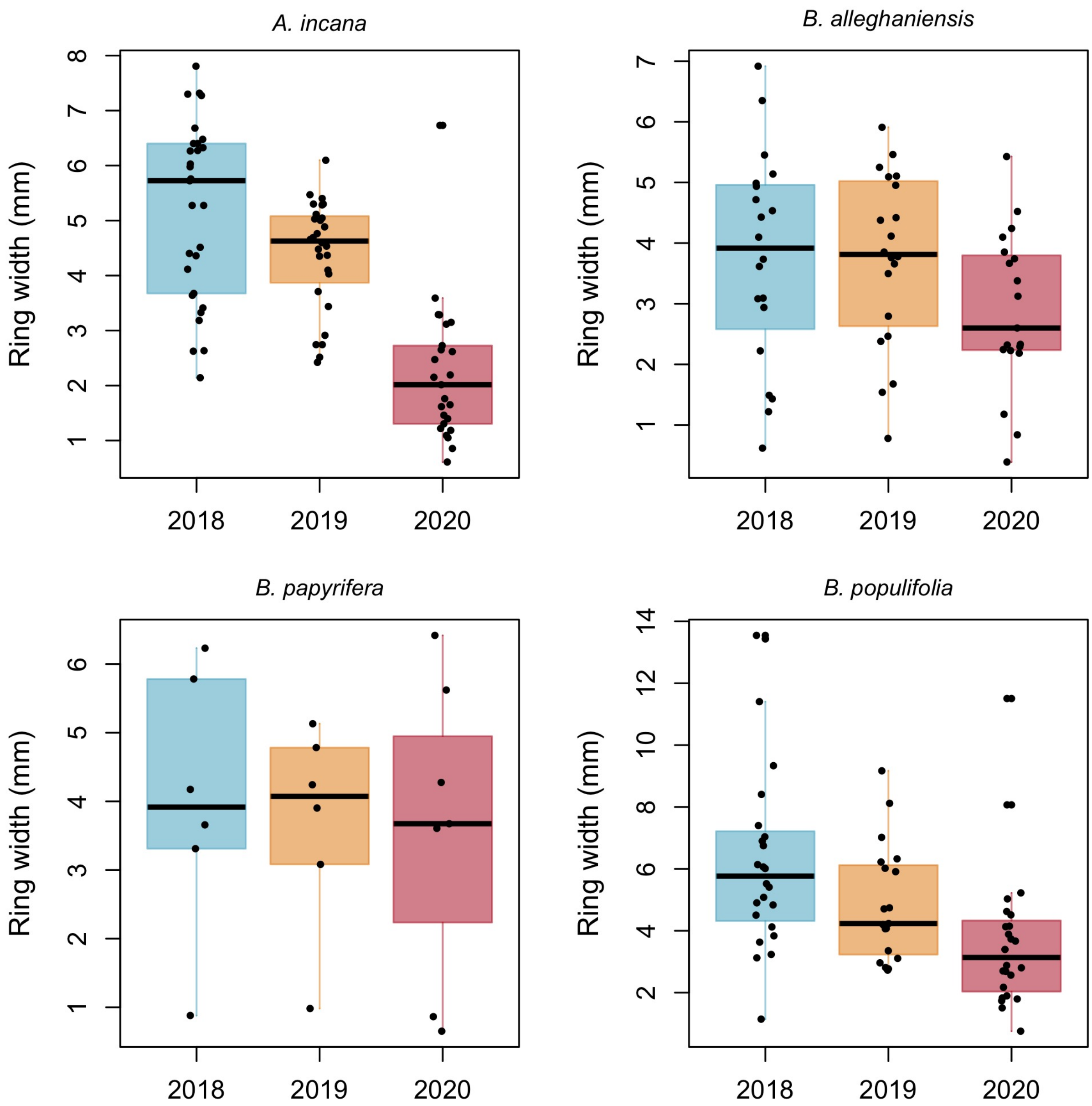

Figure S7: Boxplots representing the empirical data of our ring widths (mm) separated by species. The dots represent the raw data, the line is the mean and the box depicts the 50% UI.

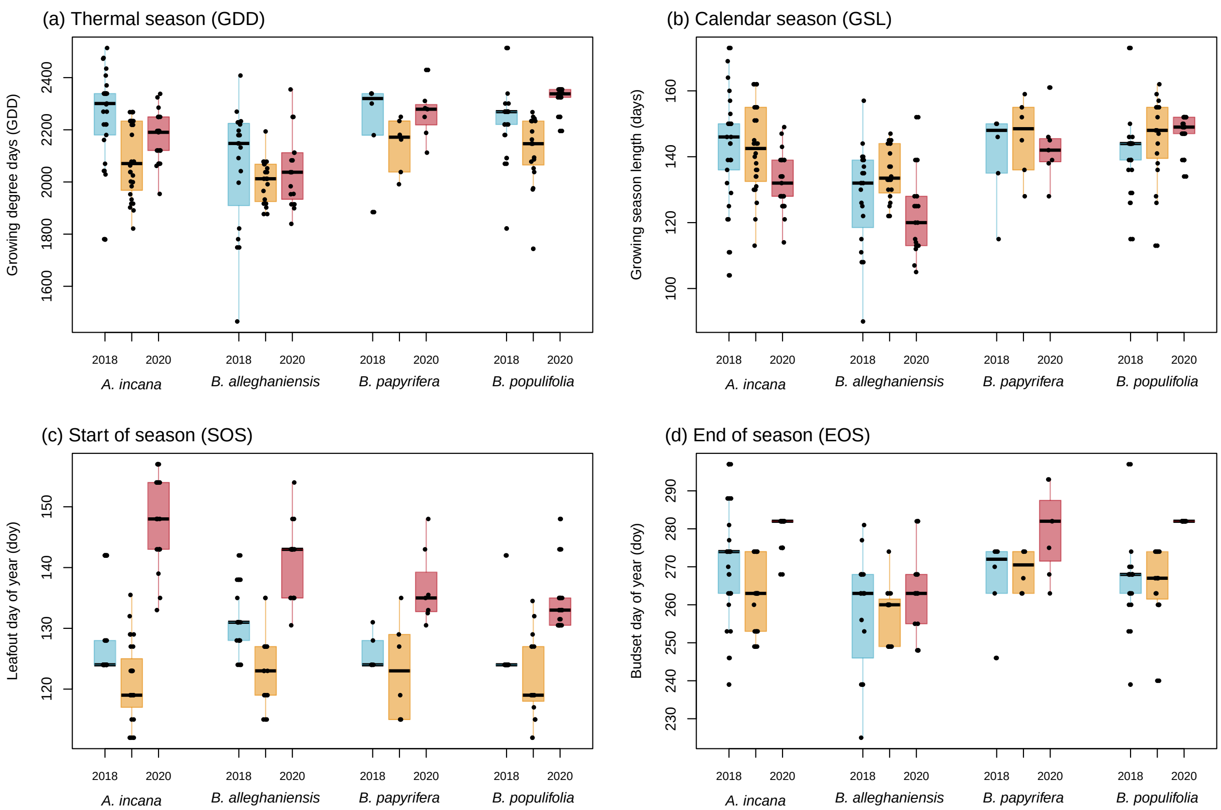

Figure S8: Boxplots representing the empirical data of all our predictors clustered by species, and for each year. The dots represent the raw data, the line is the mean and the box depicts the 50% UI.

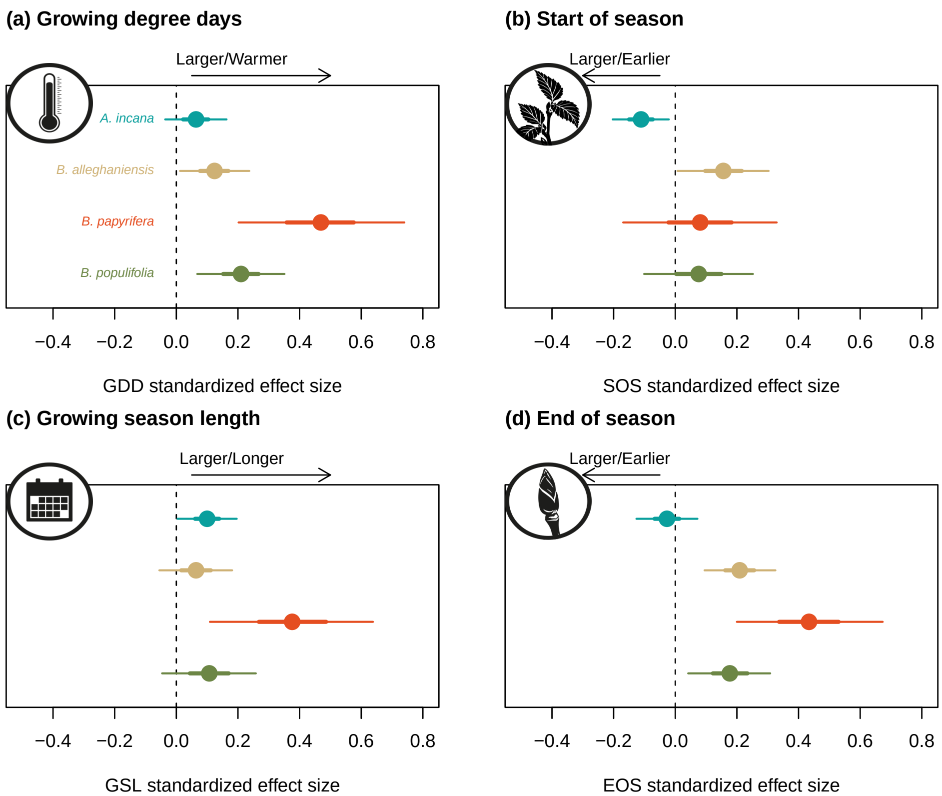

Figure S9: Effect size of each predictor across our four species. We depict ring width (log(mm)) change per adjusted predictor (z-scored; see data analyses section for more details) for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (growing degree days, growing season length, start of season and end of season).

Chapter 2

Weak response of warmer growing seasons on mature tree growth

Introduction

Anthropogenic climate change affects many different biological processes with one of the most documented being shifts in phenology—the study of recurring life history events Cleland *et al.* (2007); Menzel *et al.* (2006); Parmesan *et al.* (1999). Earlier springs and later autumns have lengthen the window of opportunity during which plants can grow (Duputié *et al.*, 2015). For trees, longer growing seasons means they should grow more which is especially significant given the critical role that forests play in offsetting human-emitted greenhouse gas emissions (Harris *et al.*, 2021; Keenan *et al.*, 2014; Stridbeck *et al.*, 2022). In temperate ecosystems, a large portion of that carbon is stored in mature, canopy trees which emphasizes the decisive role of trees at this life stage in the future of forest carbon sink (Lutz *et al.*, 2018).

However, the long standing assumption that longer growing seasons should increased growth has recently been challenged by studies that failed to find this relationship (Cabon *et al.*, 2022; Charlet De Sauvage *et al.*, 2022; Dow *et al.*, 2022; Silvestro *et al.*, 2023; Wang *et al.*, 2025). Warmer springs—that largely contribute to extending the growing season—have not increased the maximum tree growth rate or annual increment in two temperate mixed forest communities (Dow *et al.*, 2022). Increased tree growth with longer growing season is an important aspect of global vegetation models which are crucial tools to predict future climate change and guide climate policies (Friedlingstein *et al.*, 2026; Peters *et al.*, 2020). This brings forward the necessity to understand why trees may not grow more with longer growing seasons.

Wood anatomy and tree growth strategies may guide tree species responses to growing seasons. Wood anatomy in angiosperm trees impacts their growth temporality with ring-porous species starting growing several days or weeks earlier than diffuse-porous species (D'Orangeville *et al.*, 2022; Hessl *et al.*, 2026; Savage *et al.*, 2022). As a result, species with different wood anatomy experience considerably different temperature regime during the growing season. This results in an asymmetric effect of the season length on growth across species with different wood anatomy. Further, longer growing seasons may benefit slower growth species because appropriate conditions would allow them to keep their low growth rate active for a longer period (Cuny *et al.*, 2012). In contrast, faster growing species can complete their season growth increment more quickly, making extra growing days less important (Cuny *et al.*, 2012). Additionally, how trees store the carbon they accumulate (whether it is through a direct investment in growth or for long-term storage) could mute short term responses (Hessl *et al.*, 2026). These different carbon allocation strategies highlight the importance of considering the GS conditions of more than one year.

The previous year thermal conditions can affect the following year's growth. Deciduous trees often use stored carbon at the beginning of the growing season because they need to grow new leaves before starting photosynthesis (Monson *et al.*, 2018; Wolkovich *et al.*, 2025). Mature trees in particular can rely on stored carbon for their early season growth that was stored several years before (Hessl *et al.*, 2026). This shows that while the environmental conditions during the current year can have a direct effect on growth, it is likely that this poorly understood carbon allocation strategy to storage may be blurred by a multi-year lag effect.

Here, we will look at the growth and growing season length relationship of mature trees in an urban arboretum. For this, we used 11 species of mature trees across nine years and looked at the growing season length and thermal conditions during that period. We also investigated the effect of the previous year thermal conditions on the current year growth. We used growth with tree rings and leaf phenology for the start and end of season that citizen scientists collected.

Methods

Citizen science program

The Tree Spotters is a citizen science program started in 2015 that aims to train citizen scientists for accurate and rigorous phenological monitoring at the Arnold Arboretum of Harvard University (42.30 °N, -71.12 °W; Figure S1). From 2015 to 2024, hundreds of citizen scientists monitored 77 individual trees and shrubs of 14 species. They regularly followed individuals from budburst in the spring to leaf coloring in the fall using the National Phenology Network (NPN) phenophases (Denny *et al.*, 2014): leaves (483), colored leaves (498), fruits (516), ripe fruits (390), falling leaves(471), recent fruit or seed drop (504), increasing leaf size (467), breaking leaf buds (371), flowers or flower buds (500), open flowers (501), pollen release (502). Not all phenophases were recorded for every tree, for every year, and some trees are missing several years of data.

Here we use leafout to colored leaves (or leaf coloring) to define the growing season length. Taken together, our dataset comprises 321 leafout and 284 colored leaves observations (which we refer to as the 'full datasets'), but we did not have both observations for all trees in all years, so our total of matched full growing seasons duration (which we refer to as the 'restricted dataset') is 278 observations. We visually checked for SOS, EOS and GSL outliers and verified that there was no observation bias potentially caused by a particular observer or in a particular year.

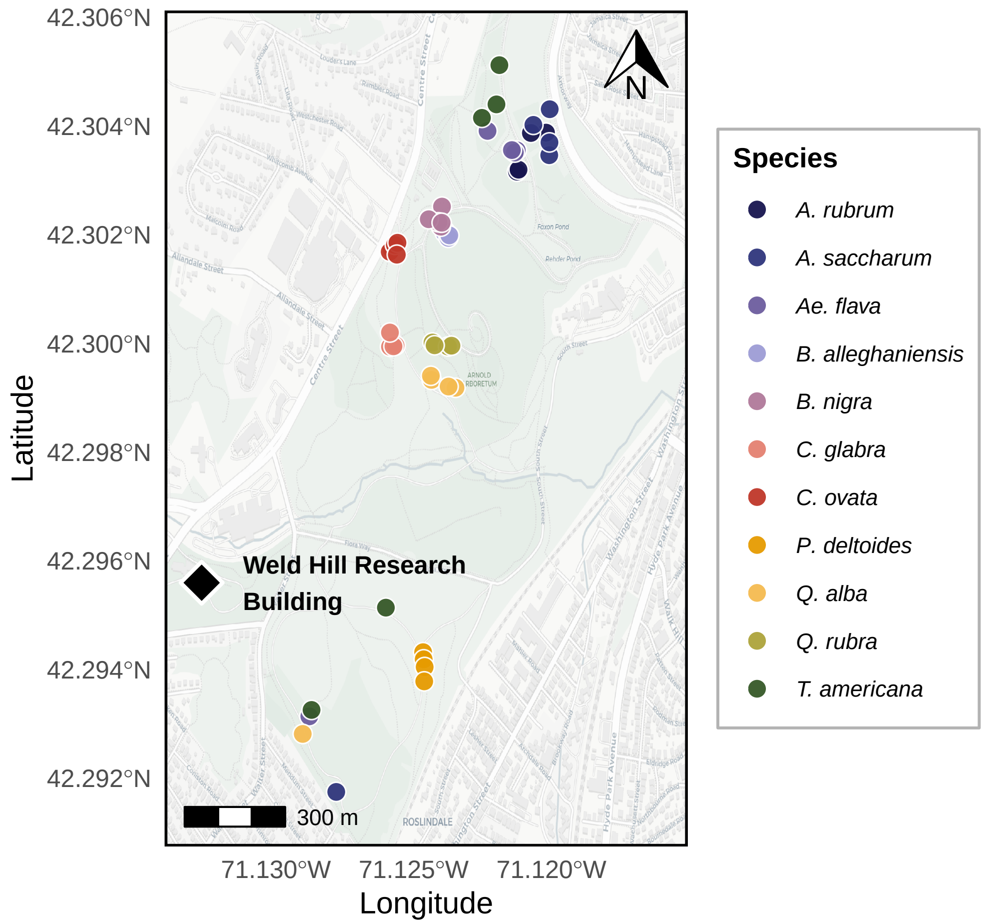

Figure S1: Map of the Arnold Arboretum of Harvard University with the locations of the trees depicted as the dots. The colors of the dots match the model estimates of model estimate figures (e.g. Figure S2). We also show the location of the Weld Hill Research Building (black diamond), where the common garden from Chapter 1 took place.

Phenological monitoring and sample collection

From 20 to 22 April 2025, we collected tree ring samples only for the tree species that were previously monitored for phenology—excluding *Fagus grandifolia* because of beach bark disease concerns—for a total of 49 individuals (Table S1). Thus, we sampled four or five individuals for 11 species: Red maple (*Acer rubrum*), Sugar maple (*Acer saccharum*), Yellow buckeye (*Aesculus flava*), Yellow birch (*Betula alleghaniensis*), River birch (*Betula nigra*), Pignut hickory (*Carya glabra*), Shagbark hickory (*Carya ovata*), Eastern cottonwood (*Populus deltoides*), White oak (*Quercus alba*), Northern red oak (*Quercus rubra*) and American basswood (*Tilia americana*). For this, we collected two 5-mm diameter cores, 15-cm length at 1.3 meters above ground, perpendicular to the slope and at 180 degrees from each other, using an increment borer (Mora Borer, Haglöfs Sweden, Bromma, Stockholm, Sweden). We cleaned the increment borer with alcohol (70% ethanol) and the inside with a brush before collecting each core. We stored the cores in modified plastic straws to promote aeration and left them to dry at ambient temperature for three months.

Sample processing, imaging and measuring

We used wooden mounts to stabilize the cores that were then sanded using progressively finer sandpaper grit (from 150 to 1000) by increments of roughly 150. We then used a high-resolution scanner to scan the cores at a resolution of 6250 dpi (Fong *et al.*, 2026), and used those images to measure the ring widths using ImageJ (Schneider *et al.*, 2012).

We visually cross-dated the ring width chronologies by assigning a calendar year to each ring and comparing annual variation of trends across the different individuals of the same species (Cook & Kairiukstis, 1990). Our short chronologies and low species replication prevented us from performing statistical cross-dating. We also did not detrend our chronologies because the ring widths during our study period (nine years) showed no consistent allometric trend (Figure S1). We included year in our models to control for effects of different years.

Data analyses

We used Bayesian hierarchical models to understand how growth shifted with four different season metrics: GDD, GSL, SOS and EOS. We used growing degree days (GDD; mean daily temperature above 5°C) between leafout and leaf coloring to calculate the thermal season and the number of days between leafout and leaf coloring for the calendar season (growing season length; GSL). Then, we used leafout for the start of season (SOS) and leaf coloring for the end of season (EOS). To better illustrate ring width responses to our season metrics, we report ring width change (log(mm)) in response to the number of GDDs accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD) for the thermal season. Similarly, we report ring width change per week of season length, leafout and leaf coloring for GSL, SOS and EOS, respectively. We report effects when the 90% UI do not overlap with zero across all models.

We combined daily climate data from two sources located next to the Arboretum for the duration of our study (2016-2024). From 2016-2019, we used the Weld Hill research station climate station (Arnold Arboretum, 2026) and the "Boston (Weld Hill)" station for the remaining years (2020-2024; Carroll *et al.*, 2026). We determined drought occurrence at the arboretum using the North American Drought Monitor Maps and used the proposed drought index to qualify their intensity (National Centers for Environmental Information (NCEI), 2026).

Thus for each season predictor, we estimated its effect on ring width (i, denoting each observation) observed for each tree (*treeid*) in each year (*year*) as:

$$\begin{aligned} ln(ringwidth)_i &\sim Normal(\mu, \sigma_y) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\ \alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha\,\text{treeid}}) \end{aligned} \tag{2.1}$$

With this model, we estimated a species-specific effect of each season predictor on growth ($\beta_{species}$), but we also controlled for variation due to species ($\alpha_{species}$), individual tree (α_{treeid}) and year (α_{year}). We first ran the models using natural units and then scaled the predictors through *z*-scoring so we could directly compare their effect sizes (Gelman & Hill, 2006). We report estimates as means and 90% uncertainty intervals (UI). We also fit the models for SOS and EOS using the full vs restricted datasets. Finally, because the wood growth onset is synchronized with a different leaf phenophase for species with different wood anatomy (ring-porous being more closely synchronized with budburst and diffuse-porous, with leafout), we ran the models using budburst instead of leafout as the SOS. This allowed us to test whether our model responses diverged across diffuse vs ring porous species (Table S1).

We used an additional model to account for the previous year thermal conditions. For this, we used the GDD accumulated between leafout and leaf coloring from the previous year (e.g. for growth year of 2020, we used the GDD of 2019 as the previous year GDD), alongside the GDD accumulated during the current year. For interpretability, we also report ring width change (log(mm)) as the number of GDDs (for both years) accumulated in an average week (seven days) of May (day of year 121 to 150 from 2016 to 2024; 58.32 GDD). Thus, we estimated the previous and current year thermal season effect on the current year's growth as:

$$\begin{aligned} \ln(\text{ringwidth})_i &\sim Normal(\mu, \sigma) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} \\ &\quad + \beta^{\text{current}}_{\text{species}} \times \text{GDD}_{\text{current}} + \beta^{\text{previous}}_{\text{species}} \times \text{GDD}_{\text{previous}} \\ \alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned} \tag{2.2}$$

While controlling for the same variation across species, individual tree and year, this model allowed us to account for the effect of the thermal season during the current year ($\beta^{\text{current}}_{\text{species}}$), but also during the previous year ($\beta^{\text{previous}}_{\text{species}}$).

Finally, we used a third Bayesian hierarchical model to determine whether the SOS and EOS of our species were related during the current year. For this, we estimated the effect of the SOS on the EOS on each tree, in each year as:

$$\begin{aligned} \text{EOS}_i &\sim Normal(\mu_i, \sigma_y) \\ \mu_i &= \alpha + \alpha_{\text{species}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{\text{species}} \times \text{SOS} \\ \alpha_{\text{treeid}} &\sim Normal(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned} \tag{2.3}$$

Results

Our dataset comprised 11 deciduous tree species with leaf phenology data spanning nine years (2016 to 2024) for a total of 321 leafout and 284 leaf coloring observations, yielding 278 complete growing seasons (GS) for 49 individuals. Throughout the duration of the study, mean summer temperatures spanned 21.53°C (2017) to 23.07°C (2020), with an average of 22.51°C. 2020 and 2022 had moderate droughts in the summer (June, July and August), which ramped up to severe in September. Additionally, a moderate drought in June 2016, ramped up to severe in July, then extreme in August and September. The calendar season (GSL) ranged from 77 days (*A. flava* in 2018) to 194 days (*A. saccharum* in 2021), while the thermal season (GDD) ranged from 1048.8 GDD (*A. flava* in 2020) to 2634.7 GDD (*T. americana* in 2016). The earliest start of season (leafout) was 08-Apr (*A. flava* in 2021) and the latest, 15-May (*C. ovata* in 2020), while the earliest leaf coloring day of year was 15-Jul (*A. flava* in 2021—same individual with earliest leafout) and the latest, 06-Nov (*T. americana* in 2016). Annual ring widths across years varied from 0.43 mm (*A. saccharum* in 2019) to 11.75 mm (*T. americana* in 2019; Figure S3).

Across our eleven species and season predictors, we did not find any clear response of the growing season length on growth. We report estimates as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henceforth 'RW/day' or 'RW/GDD' with days and GDD being adjusted to represent more interpretable units (see 'Data analyses' section in Methods for more details on predictor adjustment). Longer thermal season did not increase growth for any species and decreased growth for *B. alleghaniensis* (β = -0.08, UI: -0.12, -0.03 RW/GDD; Figure S2a and e; Table S2). Further, a longer calendar season did not change growth for any species (Figure S2b and f; Table S2). On a scaled axis (through z-scoring, Gelman & Hill, 2006), the thermal and calendar seasons were similarly predictive (see Figure S11.

Previous year climate also had limited effects on growth. For the one species that showed decreased growth with warmer thermal season, this effect persisted for the current year, but previous year GDD actually increased growth for *B. alleghaniensis* (β = 0.15, UI: 0.02, 0.28 RW/GDD; Figure S3b; Table S5). *B. alleghaniensis*'s congeneric, *B. nigra*, in contrast grew less with previous year GDD (β = -0.08, UI: -0.17, -0.00 RW/GDD; Figure S3b; Table S5) and current year GDD increased growth for that species (β = 0.12, UI: 0.03, 0.21 RW/GDD; Figure S3a; Table S5). Interestingly, *B. nigra* did not solely respond to current year's conditions (Figure S2a and e; Table S2), but grew more with warmer thermal season only when the previous conditions were considered.

An earlier start of season did not increase growth for any species and decreased growth for *T. americana* ($\beta = 0.15$, UI: 0.02, 0.27 RW/day; Figure S2c and g; Table S2), which did not respond to the thermal or the calendar season. Additionally, a later end of season did not increase growth for any species and decreased growth for *B. alleghaniensis* (β = -0.09, UI: -0.16, -0.01 RW/day; Figure S2d and h; Table S2), which also decreased growth with warmer thermal season. Then, the estimates were unchanged when using the full or restricted datasets (SOS: $n = 321$ and EOS: $n = 284$ vs $n = 278$; Figure S9b), or when using budburst instead of leafout as the start of the growing season (Figure S10c).

Finally, a later SOS delayed EOS only for *P. deltoides* (β: 1.04, UI: 0.39, 1.70 EOS/SOS; Figure S12). We did not find any clear difference in baseline growth across years (Figure S2; Table S3) and between species (Figure S6; S7). Growth was similar across individual trees (Figure S8).

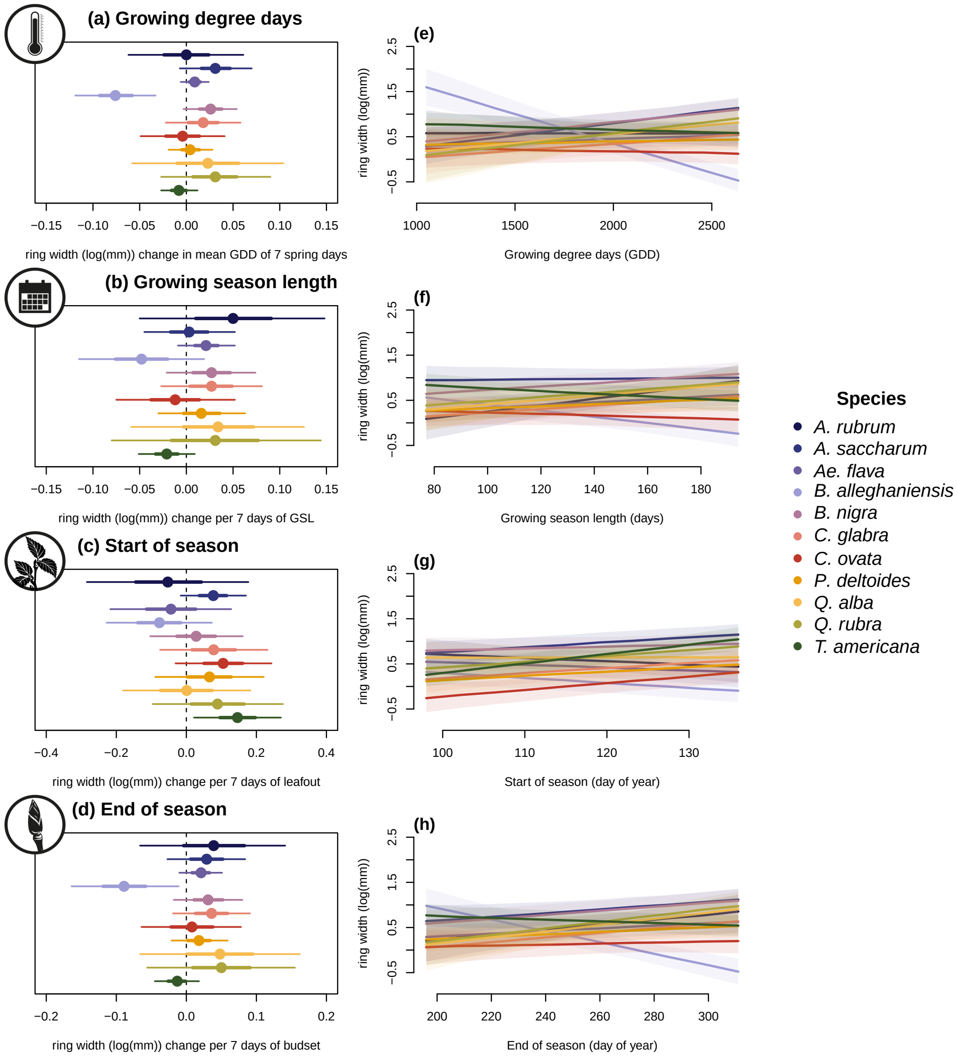

Figure S2: Estimates of ring width change (log(mm)) as a function of four different phenological predictors: growing degree days of the growing season (mean daily temperature above 5°C); growing season length; start of season; end of season across 11 species: (Red maple (*Acer rubrum*), Sugar maple (*Acer saccharum*), Yellow buckeye (*Aesculus flava*), Yellow birch (*Betula alleghaniensis*), River birch (*Betula nigra*), Pignut hickory (*Carya glabra*), Shagbark hickory (*Carya ovata*), Eastern cottonwood (*Populus deltoides*), White oak (*Quercus alba*), Northern red oak (*Quercus rubra*) and American basswood (*Tilia americana*)); species colors are the same across the panels). Panels on the left (a to d) represent the mean model slope estimates for each species (dots) with 50% and 90% uncertainty intervals (thicker and thinner lines, respectively), for each predictor, which we represent through ring width change in mm on the log scale per adjusted predictor (see 'Data analyses' section in Methods for more details on predictor adjustment). See Table S2 for parameter estimates. Panels on the right (e to h) show the ring width response curves as a function of each predictor, with the lines depicting the mean response at each predictor value, and the shaded area, the 50% uncertainty intervals. These models do not include an effect of the previous year GDD (see Figure S3 for these estimates).

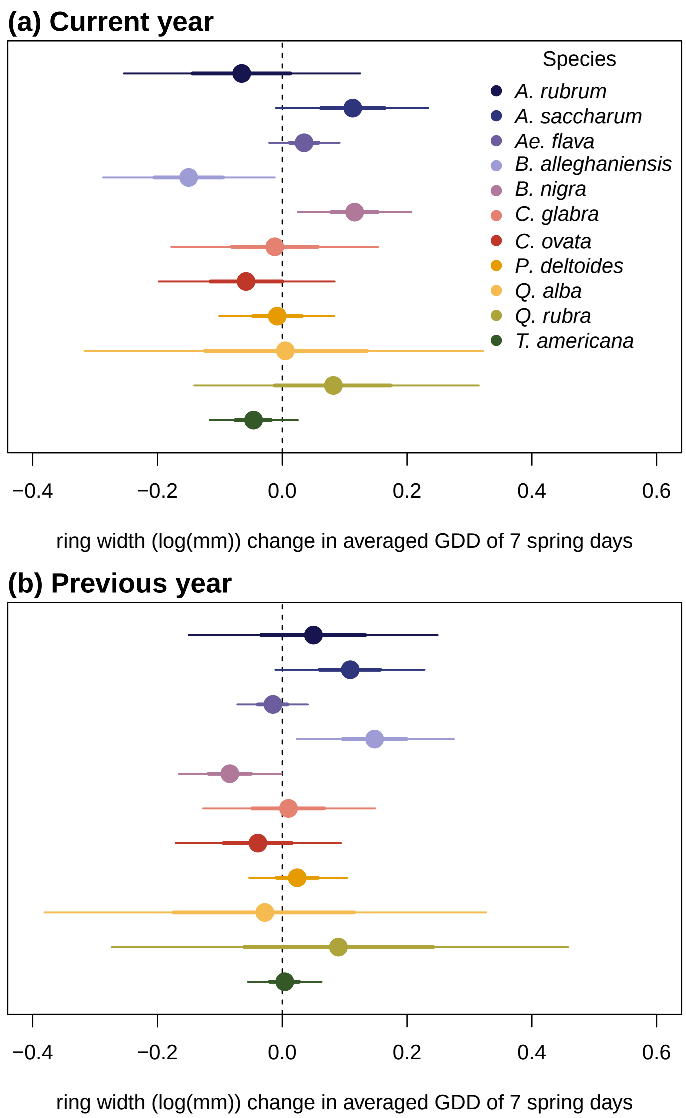

Figure S3: Model slope estimates representing ring width change (log(mm)) as a function of the current year growing degree days (GDD; a) and the previous year GDD (b). GDD represent the accumulated growing degree days (mean daily temperature above 5°C)) between leafout and leaf colouring and the models estimates shows ring width change (log(mm)) per scaled GDD (see 'Data analyses' section in Methods). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI).

Discussion

In this study, we used 49 trees of 11 species to understand if mature trees grew more with longer growing seasons (GS). To do this, we used annual growth with tree rings and leaf phenology for the SOS and EOS. Our findings reveal that warmer and longer GS did not increase growth for any species and decreased growth for one. We also found that the previous year thermal conditions shifted growth trends for two species. Finally, most species did not respond to the start and end of the GS.

While our species share an array of growth strategies (see Table S1), we did not find a consistent growth trend across them. Only *B. alleghaniensis* responded to the thermal season and its decrease in growth suggests that this species may be harmed by warmer seasons. Consequently, *B. alleghaniensis* also grew less with a later EOS which is a period characterized by fast GDD accumulation. The fact that *B. alleghaniensis* appears to be negatively impacted by warmer seasons suggest that this species could face local extirpation at its southernmost range because growth would not keep up with warmer and longer seasons (Aitken *et al.*, 2008; Doak & Morris, 2010). In addition, the fast-growing *T. americana* (Table S1; Burns *et al.*, 1990) grew less with an earlier SOS—when the days are short and cool. This is contrast with our expectation of species with this life-history strategy which should provide them an edge by starting to grow earlier in the season (Loughnan *et al.*, 2025). Ultimately, these findings suggest a weak signal of life-history strategies in growth responses to the GS length and thermal conditions. Though, we found that these trends shifted slightly when considering also the previous year thermal conditions.

The growth response of two *Betula* species shifted when we included the previous year thermal season as a driver. These two congeneric species (*B. alleghaniensis* and *B. nigra*) had diametrically opposed response trends, however. *B. alleghaniensis* growth decreased with the current year thermal conditions, but increased with a warmer previous year, possibly revealing a carbon allocation strategy that prioritize carbon storage above short-term growth. This is in contrast with the expectation from diffuse-porous species that tend to allocate less carbon to storage and more to growth (Barbaroux & Breda, 2002; Muhr *et al.*, 2016). In opposition, *B. nigra* increase in growth during the current year—only when the previous year conditions were included—confirms the expectation from diffuse-porous species of a short-term investment in growth with less allocation to storage. These opposite responses of two closely related species that have slightly different life-history strategies (Table S1; Burns *et al.*, 1990) highlight the importance of considering how fast trees grow in growth and GS length models.

Also, considering not solely the current year conditions when studying mature tree growth could help elucidate tree growth dynamics as even fast-growth species may have their responses blurred by carbon allocation to something else than growth. Our findings reveal a weak response of the GS length and conditions on growth across species, corroborating previous findings on mature trees (Delpierre *et al.*, 2016; Silvestro *et al.*, 2023; Čufar *et al.*, 2008). Most of our studied species did not grow more with a warmer thermal or longer calendar season. However, out results emphasize the importance of considering the previous year conditions in future growth and GS length studies because of the vast array of carbon allocation strategies that may blur short-term tree growth responses. GS are expected to warm and lengthen with, but it is still dubious how mature trees will track this trend. Whether tree growth stalls or decrease, both responses will influence the contribution of temperate forest to act as a global net carbon sink.

Supplemental

Supplemental Methods

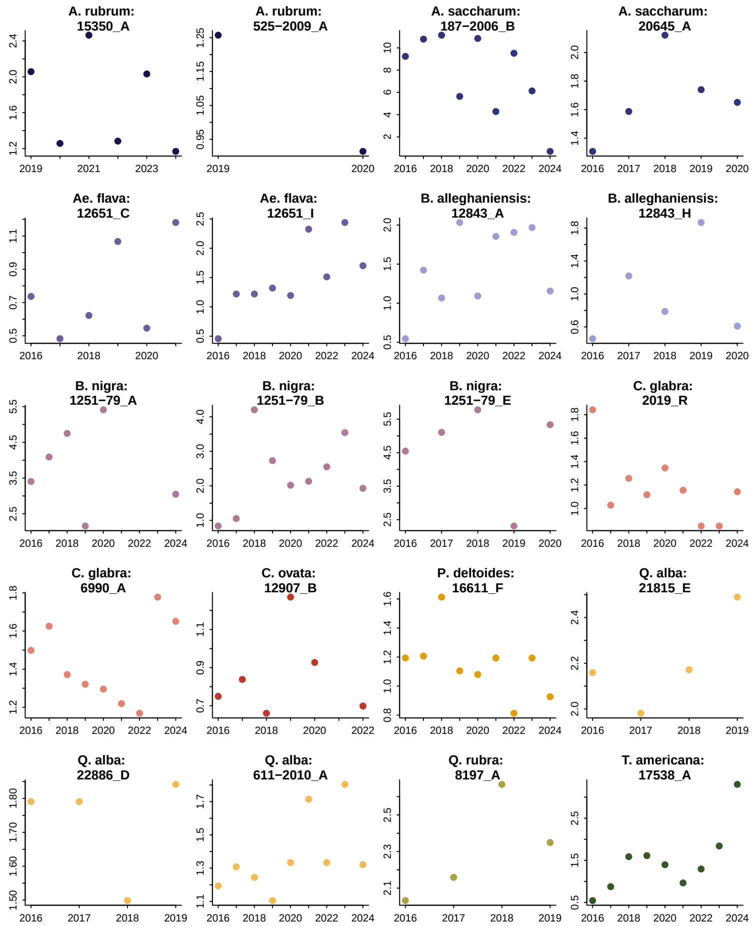

Figure S1: We randomly sampled 20 trees from our dataset and represent the ring width as a function of year to represent the allometry patterns across all years we had phenology data for.

Table S1: Treespotters species grouped by tree type, growth speed, and wood anatomy. We collected the growth speed information for each species from Burns *et al.* (1990)We sampled a total of 49 individual trees

Common Name (Latin binomial)	Growth speed	Wood Anatomy	n
American basswood (*Tilia americana*)	Fast	Diffuse-porous	5
Eastern cottonwood (*Populus deltoides*)	Fast	Diffuse-porous	4
Northern red oak (*Quercus rubra*)	Moderate	Ring-porous	4
White oak (*Quercus alba*)	Slow	Ring-porous	5
Pignut hickory (*Carya glabra*)	Slow	Ring-porous	4
Shagbark hickory (*Carya ovata*)	Moderate	Ring-porous	5
River birch (*Betula nigra*)	Fast	Diffuse-porous	4
Yellow birch (*Betula alleghaniensis*)	Moderate	Diffuse-porous	4
Sugar maple (*Acer saccharum*)	Slow	Diffuse-porous	5
Red maple (*Acer rubrum*)	Moderate	Diffuse-porous	4
Yellow buckeye (*Aesculus flava*)	Moderate	Diffuse-porous	5

Supplemental results

Table S2: Species level slope parameters which represent ring width change (log(mm)), per scaled predictor (see 'Data analyses' section in Methods for more details on predictor scaling). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). Table S3 reports all the models parameters.

Species	GDD	GSL	SOS	EOS
A. rubrum	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
A. saccharum	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
Ae. flava	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
B. alleghaniensis	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
B. nigra	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
C. glabra	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
C. ovata	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
P. deltoides	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
Q. alba	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
Q. rubra	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
T. americana	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]

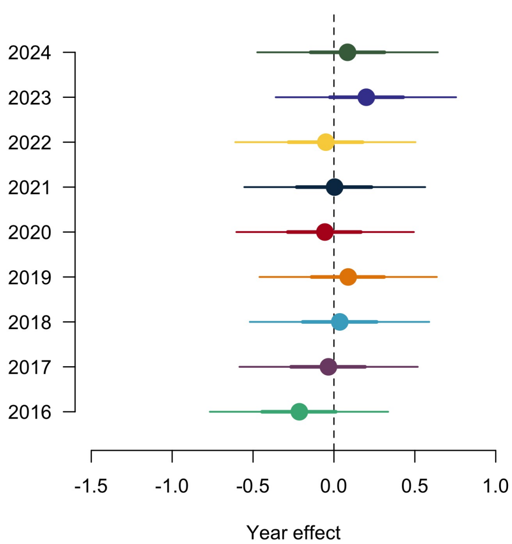

Figure S2: Ring width (log(mm)) deviations from the grand mean for each year using the GDD model (Table S3). The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

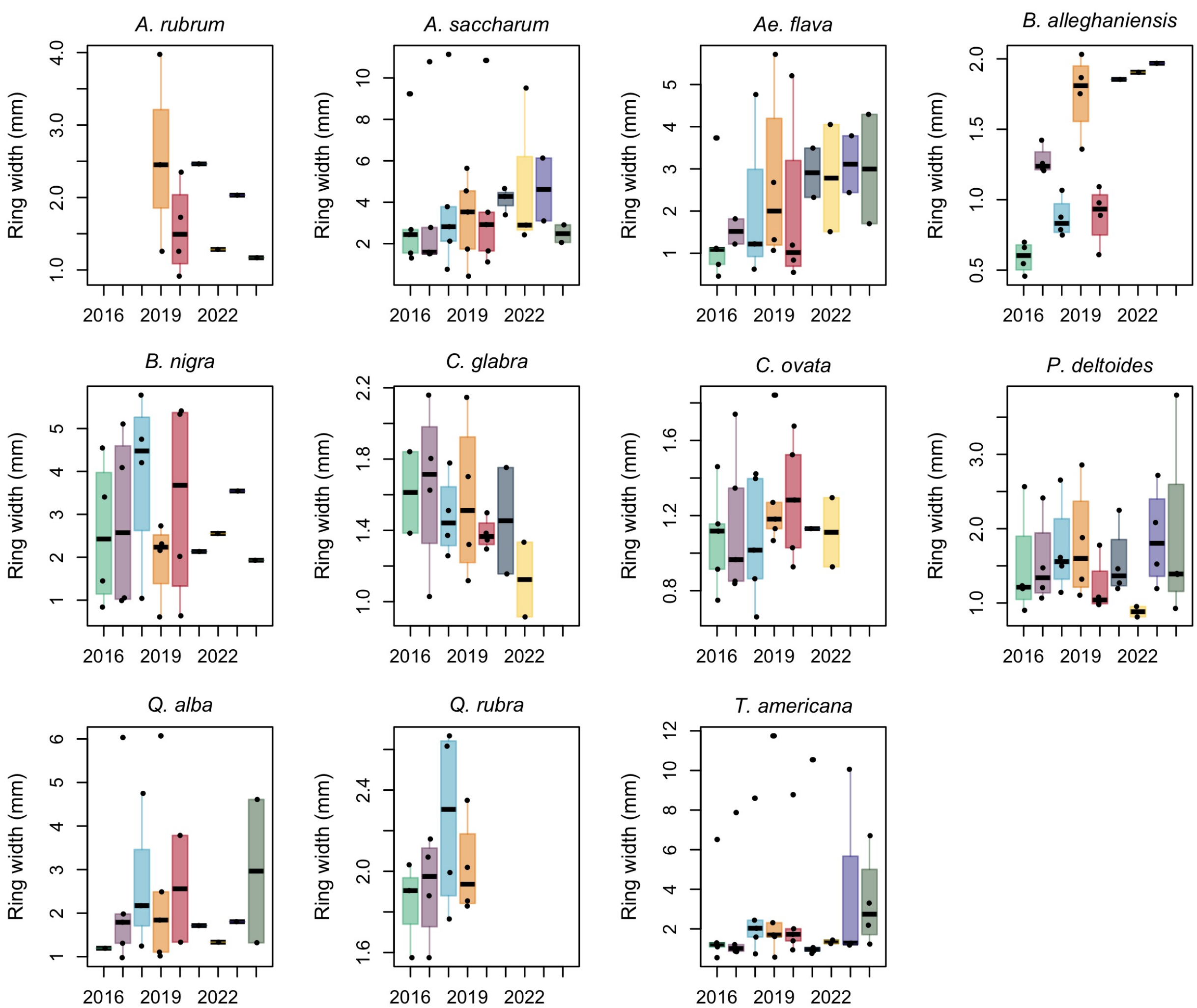

Figure S3: Boxplot representing the empirical data of our ring widths (mm) across years, facetted by species. The dots represent the raw data, the line the mean and the box the interquantile range (IQR).

Table S3: Summary output of each parameter from our four predictors, using ring width (log(mm)) as the response variable (See equation 2.1). We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season). See 'Data analyses' section in Methods for more details.

Parameter	GDD	GSL	SOS	EOS
α	0.61 [-2.08, 3.25]	0.49 [-2.09, 3.03]	0.24 [-2.39, 2.94]	0.23 [-2.48, 2.98]
$\sigma_{\alpha_{tree}}$	0.55 [0.45, 0.67]	0.55 [0.45, 0.68]	0.54 [0.44, 0.66]	0.55 [0.45, 0.67]
σ_y	0.31 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]
$\beta_{species_{A.rubrum}}$	0.00 [-0.06, 0.06]	0.05 [-0.05, 0.15]	-0.05 [-0.28, 0.18]	0.04 [-0.07, 0.14]
$\beta_{species_{A.saccharum}}$	0.03 [-0.01, 0.07]	0.00 [-0.04, 0.05]	0.08 [-0.02, 0.17]	0.03 [-0.03, 0.08]
$\beta_{species_{Ae.flava}}$	0.01 [-0.01, 0.02]	0.02 [-0.01, 0.05]	-0.04 [-0.22, 0.13]	0.02 [-0.01, 0.05]
$\beta_{species_{B.alleghaniensis}}$	-0.08 [-0.12, -0.03]	-0.05 [-0.12, 0.02]	-0.08 [-0.23, 0.07]	-0.09 [-0.16, -0.01]
$\beta_{species_{B.nigra}}$	0.03 [-0.00, 0.05]	0.03 [-0.02, 0.07]	0.03 [-0.10, 0.16]	0.03 [-0.02, 0.08]
$\beta_{species_{C.glabra}}$	0.02 [-0.02, 0.06]	0.03 [-0.03, 0.08]	0.08 [-0.07, 0.23]	0.04 [-0.02, 0.09]
$\beta_{species_{C.ovata}}$	-0.00 [-0.05, 0.04]	-0.01 [-0.07, 0.05]	0.10 [-0.03, 0.24]	0.01 [-0.06, 0.08]
$\beta_{species_{P.deltoides}}$	0.00 [-0.02, 0.03]	0.02 [-0.03, 0.06]	0.07 [-0.09, 0.22]	0.02 [-0.02, 0.06]
$\beta_{species_{Q.alba}}$	0.02 [-0.06, 0.10]	0.03 [-0.06, 0.13]	0.00 [-0.18, 0.18]	0.05 [-0.07, 0.16]
$\beta_{species_{Q.rubra}}$	0.03 [-0.03, 0.09]	0.03 [-0.08, 0.14]	0.09 [-0.10, 0.28]	0.05 [-0.06, 0.15]
$\beta_{species_{T.americana}}$	-0.01 [-0.03, 0.01]	-0.02 [-0.05, 0.01]	0.15 [0.02, 0.27]	-0.01 [-0.04, 0.02]
$\alpha_{year_{2016}}$	-0.05 [-3.40, 3.35]	-0.95 [-4.12, 2.17]	1.23 [-3.23, 5.60]	-1.08 [-5.62, 3.44]
$\alpha_{year_{2017}}$	-0.90 [-3.86, 2.10]	0.43 [-2.34, 3.17]	-0.57 [-3.59, 2.35]	-0.37 [-3.79, 3.02]
$\alpha_{year_{2018}}$	-0.49 [-3.20, 2.19]	-0.46 [-3.09, 2.14]	0.93 [-2.69, 4.47]	-0.49 [-3.33, 2.38]
$\alpha_{year_{2019}}$	2.32 [-0.68, 5.34]	0.61 [-2.29, 3.46]	1.17 [-2.34, 4.58]	3.25 [-0.47, 6.96]
$\alpha_{year_{2020}}$	-0.68 [-3.54, 2.12]	-0.13 [-2.82, 2.58]	0.17 [-3.18, 3.40]	-0.50 [-3.65, 2.67]
$\alpha_{year_{2021}}$	-0.89 [-3.89, 2.06]	-0.64 [-3.44, 2.12]	-1.17 [-4.65, 2.33]	-1.16 [-4.46, 2.12]
$\alpha_{year_{2022}}$	-0.32 [-3.35, 2.74]	-0.07 [-2.88, 2.73]	-1.96 [-5.42, 1.40]	-0.35 [-4.01, 3.29]
$\alpha_{year_{2023}}$	-0.38 [-3.14, 2.40]	-0.39 [-3.09, 2.26]	-1.04 [-4.60, 2.52]	-0.49 [-3.54, 2.50]
$\alpha_{year_{2024}}$	-0.85 [-4.70, 3.03]	-0.56 [-3.66, 2.52]	0.40 [-3.47, 4.32]	-1.44 [-6.32, 3.48]
$\alpha_{species_{A.rubrum}}$	-1.09 [-4.47, 2.23]	-0.43 [-3.80, 2.96]	-1.08 [-4.98, 2.91]	-1.44 [-6.04, 3.22]
$\alpha_{species_{A.saccharum}}$	0.30 [-2.41, 3.02]	0.59 [-1.98, 3.18]	-2.02 [-5.24, 1.14]	0.92 [-1.93, 3.81]
$\alpha_{species_{Ae.flava}}$	-0.21 [-0.77, 0.34]	-0.22 [-0.76, 0.35]	-0.25 [-0.80, 0.31]	-0.25 [-0.81, 0.29]
$\alpha_{species_{B.alleghaniensis}}$	-0.03 [-0.58, 0.52]	-0.06 [-0.59, 0.52]	0.00 [-0.54, 0.56]	-0.07 [-0.63, 0.48]
$\alpha_{species_{B.nigra}}$	0.04 [-0.52, 0.59]	0.02 [-0.52, 0.59]	-0.00 [-0.55, 0.56]	0.01 [-0.54, 0.56]
$\alpha_{species_{C.glabra}}$	0.09 [-0.46, 0.64]	0.06 [-0.48, 0.63]	0.07 [-0.47, 0.64]	0.06 [-0.50, 0.62]
$\alpha_{species_{C.ovata}}$	-0.06 [-0.60, 0.49]	-0.07 [-0.61, 0.50]	-0.12 [-0.66, 0.45]	-0.07 [-0.63, 0.47]
$\alpha_{species_{P.deltoides}}$	0.00 [-0.55, 0.56]	-0.00 [-0.55, 0.57]	0.01 [-0.54, 0.58]	-0.01 [-0.58, 0.54]
$\alpha_{species_{Q.alba}}$	-0.05 [-0.61, 0.50]	-0.07 [-0.63, 0.50]	-0.07 [-0.63, 0.49]	-0.08 [-0.65, 0.48]
$\alpha_{species_{Q.rubra}}$	0.20 [-0.36, 0.76]	0.18 [-0.38, 0.75]	0.20 [-0.35, 0.78]	0.17 [-0.40, 0.72]
$\alpha_{species_{T.americana}}$	0.08 [-0.47, 0.64]	0.07 [-0.47, 0.66]	0.07 [-0.48, 0.64]	0.08 [-0.49, 0.64]

Table S4: Summary output of each parameter from the previous year model (Equation 2.2) using ring width (log(mm)) as the response variable. We report estimates as mean and 90% uncertainty intervals (in square brackets). see 'Data analyses' section in Methods for more details

Parameter	Estimate
α	0.20 [-1.10, 1.48]
$\sigma_{\alpha_{tree}}$	0.58 [0.47, 0.72]
σ_y	0.29 [0.26, 0.32]
$\beta^{\text{current}}_{\text{species}_{A.rubrum}}$	-0.07 [-0.25, 0.12]
$\beta^{\text{current}}_{\text{species}_{A.saccharum}}$	0.11 [-0.01, 0.23]
$\beta^{\text{current}}_{\text{species}_{Ae.flava}}$	0.04 [-0.02, 0.09]
$\beta^{\text{current}}_{\text{species}_{B.alleghaniensis}}$	-0.15 [-0.29, -0.01]
$\beta^{\text{current}}_{\text{species}_{B.nigra}}$	0.12 [0.03, 0.21]
$\beta^{\text{current}}_{\text{species}_{C.glabra}}$	-0.01 [-0.18, 0.15]
$\beta^{\text{current}}_{\text{species}_{C.ovata}}$	-0.06 [-0.20, 0.08]
$\beta^{\text{current}}_{\text{species}_{P.deltoides}}$	-0.01 [-0.10, 0.08]
$\beta^{\text{current}}_{\text{species}_{Q.alba}}$	0.01 [-0.32, 0.32]
$\beta^{\text{current}}_{\text{species}_{Q.rubra}}$	0.08 [-0.14, 0.32]
$\beta^{\text{current}}_{\text{species}_{T.americana}}$	-0.05 [-0.12, 0.03]
$\beta^{\text{previous}}_{\text{species}_{A.rubrum}}$	0.05 [-0.15, 0.25]
$\beta^{\text{previous}}_{\text{species}_{A.saccharum}}$	0.11 [-0.01, 0.23]
$\beta^{\text{previous}}_{\text{species}_{Ae.flava}}$	-0.01 [-0.07, 0.04]
$\beta^{\text{previous}}_{\text{species}_{B.alleghaniensis}}$	0.15 [0.02, 0.28]
$\beta^{\text{previous}}_{\text{species}_{B.nigra}}$	-0.08 [-0.17, -0.00]
$\beta^{\text{previous}}_{\text{species}_{C.glabra}}$	0.01 [-0.13, 0.15]
$\beta^{\text{previous}}_{\text{species}_{C.ovata}}$	-0.04 [-0.17, 0.09]
$\beta^{\text{previous}}_{\text{species}_{P.deltoides}}$	0.02 [-0.05, 0.10]
$\beta^{\text{previous}}_{\text{species}_{Q.alba}}$	-0.03 [-0.38, 0.33]
$\beta^{\text{previous}}_{\text{species}_{Q.rubra}}$	0.09 [-0.27, 0.46]
$\beta^{\text{previous}}_{\text{species}_{T.americana}}$	0.00 [-0.06, 0.06]
$\alpha_{\text{year}_{2016}}$	0.44 [-3.10, 3.90]
$\alpha_{\text{year}_{2017}}$	-2.20 [-4.79, 0.36]
$\alpha_{\text{year}_{2018}}$	0.04 [-1.53, 1.59]
$\alpha_{\text{year}_{2019}}$	-0.06 [-2.61, 2.44]
$\alpha_{\text{year}_{2020}}$	0.30 [-1.39, 1.99]
$\alpha_{\text{year}_{2021}}$	0.18 [-1.77, 2.13]
$\alpha_{\text{year}_{2022}}$	1.16 [-1.18, 3.47]
$\alpha_{\text{year}_{2023}}$	-0.01 [-1.92, 1.90]
$\alpha_{\text{year}_{2024}}$	0.97 [-5.55, 7.68]
$\alpha_{\text{species}_{A.rubrum}}$	-1.73 [-8.12, 4.60]
$\alpha_{\text{species}_{A.saccharum}}$	0.93 [-0.65, 2.50]
$\alpha_{\text{species}_{Ae.flava}}$	0.07 [-0.48, 0.62]
$\alpha_{\text{species}_{B.alleghaniensis}}$	0.06 [-0.50, 0.61]
$\alpha_{\text{species}_{B.nigra}}$	-0.30 [-1.02, 0.42]
$\alpha_{\text{species}_{C.glabra}}$	-0.04 [-0.58, 0.53]
$\alpha_{\text{species}_{C.ovata}}$	0.00 [-0.55, 0.56]
$\alpha_{\text{species}_{P.deltoides}}$	-0.00 [-0.56, 0.56]
$\alpha_{\text{species}_{Q.alba}}$	0.17 [-0.40, 0.73]

Table S5: Summary output of each parameter from the model on the effect of SOS (days) on EOS (days; Equation 2.3). We report estimates as mean and 90% uncertainty intervals (in square brackets). see 'Data analyses' section in Methods for more details

Parameter	Estimate
α	29.51 [23.54, 35.39]
$\sigma_{\alpha_{treeid}}$	1.18 [0.83, 1.56]
σ_y	2.06 [1.90, 2.23]
$\beta_{\text{species}_{A.rubrum}}$	0.63 [-0.14, 1.42]
$\beta_{\text{species}_{A.saccharum}}$	0.18 [-0.29, 0.66]
$\beta_{\text{species}_{Ae.flava}}$	0.38 [-0.32, 1.08]
$\beta_{\text{species}_{B.alleghaniensis}}$	0.50 [-0.15, 1.16]
$\beta_{\text{species}_{B.nigra}}$	0.44 [-0.16, 1.06]
$\beta_{\text{species}_{C.glabra}}$	0.48 [-0.17, 1.15]
$\beta_{\text{species}_{C.ovata}}$	0.40 [-0.22, 1.01]
$\beta_{\text{species}_{P.deltoides}}$	1.04 [0.39, 1.70]
$\beta_{\text{species}_{Q.alba}}$	0.35 [-0.34, 1.04]
$\beta_{\text{species}_{Q.rubra}}$	0.50 [-0.19, 1.22]
$\beta_{\text{species}_{T.americana}}$	0.46 [-0.14, 1.05]
$\alpha_{\text{year}_{2016}}$	-2.26 [-15.00, 10.18]
$\alpha_{\text{year}_{2017}}$	7.21 [-1.83, 16.23]
$\alpha_{\text{year}_{2018}}$	0.36 [-10.75, 11.25]
$\alpha_{\text{year}_{2019}}$	-0.23 [-11.38, 10.91]
$\alpha_{\text{year}_{2020}}$	0.27 [-10.43, 10.84]
$\alpha_{\text{year}_{2021}}$	-0.20 [-11.26, 10.69]
$\alpha_{\text{year}_{2022}}$	1.92 [-9.19, 13.03]
$\alpha_{\text{year}_{2023}}$	-11.53 [-23.17, 0.25]
$\alpha_{\text{year}_{2024}}$	3.61 [-8.48, 15.48]
$\alpha_{\text{species}_{A.rubrum}}$	0.14 [-12.16, 12.19]
$\alpha_{\text{species}_{A.saccharum}}$	-0.27 [-10.42, 9.76]
$\alpha_{\text{species}_{Ae.flava}}$	1.86 [-0.38, 4.14]
$\alpha_{\text{species}_{B.alleghaniensis}}$	1.17 [-1.06, 3.41]
$\alpha_{\text{species}_{B.nigra}}$	0.47 [-1.80, 2.72]
$\alpha_{\text{species}_{C.glabra}}$	-0.42 [-2.70, 1.80]
$\alpha_{\text{species}_{C.ovata}}$	-1.02 [-3.27, 1.27]
$\alpha_{\text{species}_{P.deltoides}}$	-0.23 [-2.54, 2.06]
$\alpha_{\text{species}_{Q.alba}}$	-0.87 [-3.21, 1.45]
$\alpha_{\text{species}_{Q.rubra}}$	-0.28 [-2.63, 2.06]
$\alpha_{\text{species}_{T.americana}}$	-1.50 [-3.85, 0.81]

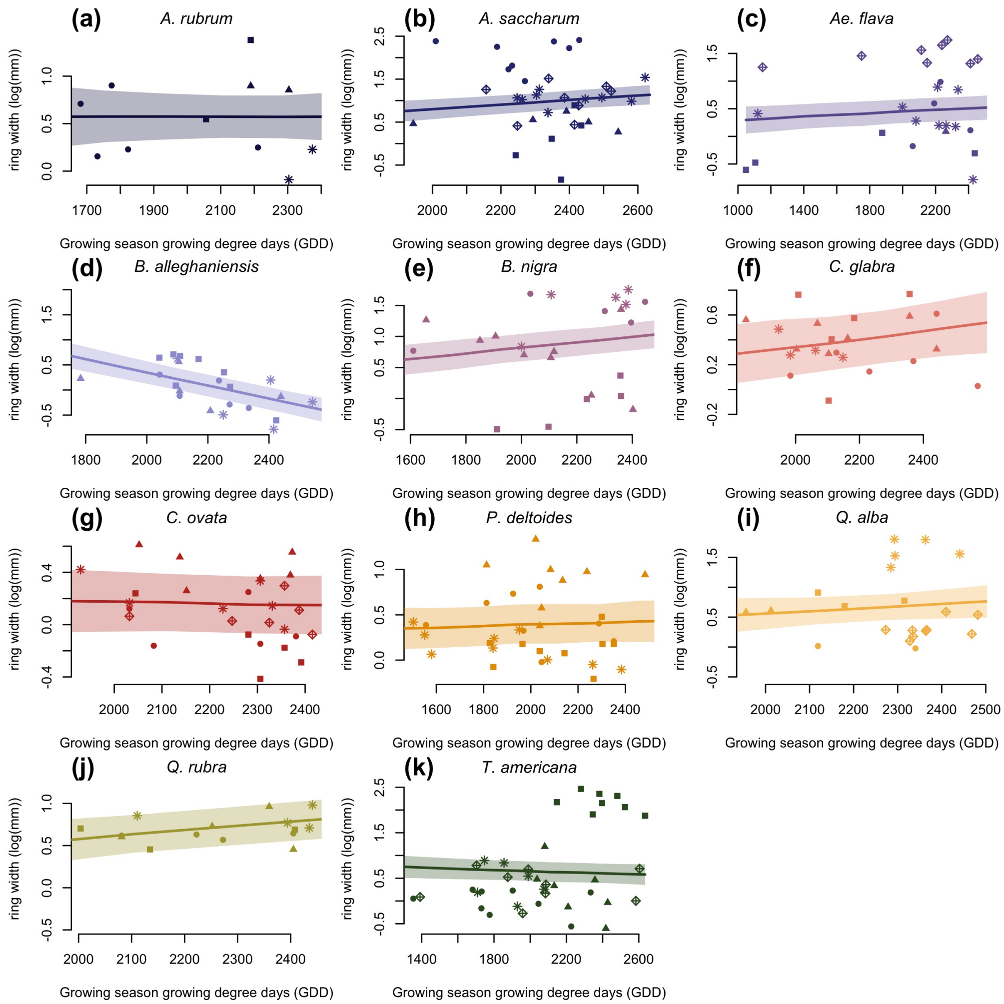

Figure S4: Ring width (log(mm)) trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

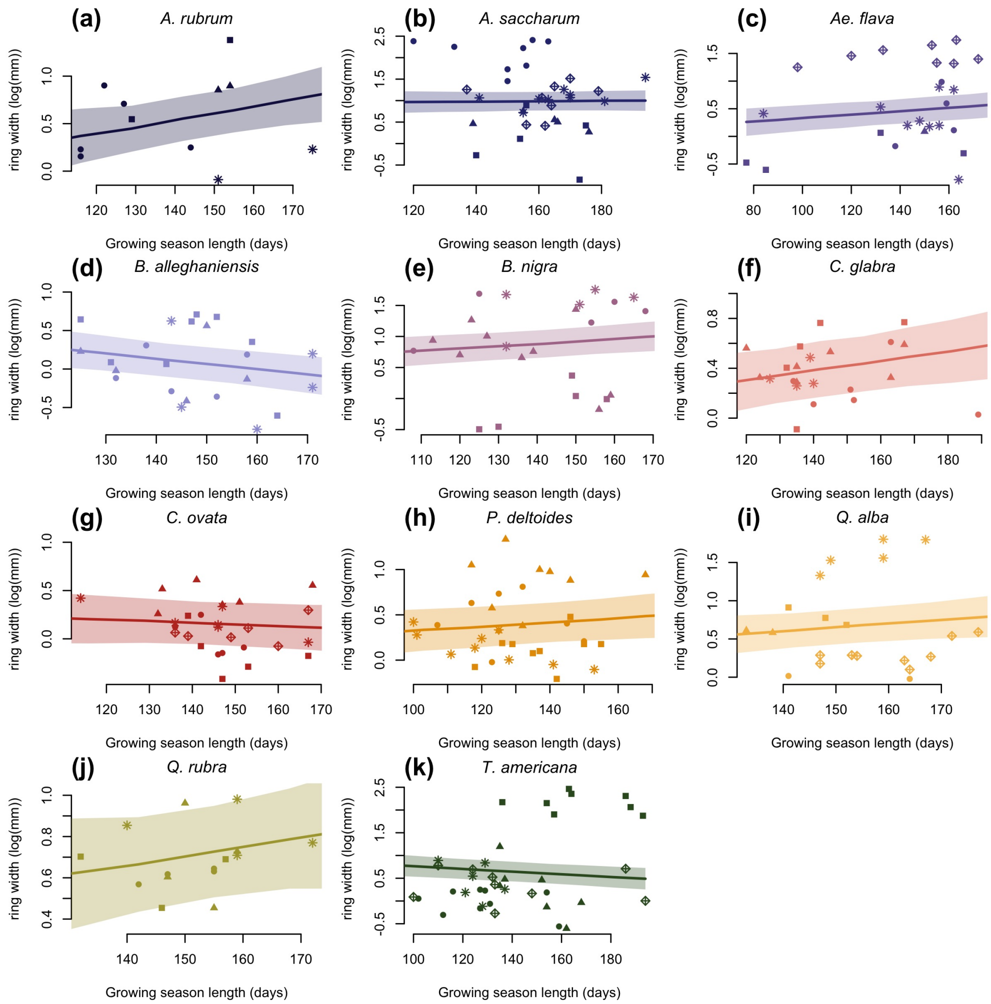

Figure S5: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by treeids.

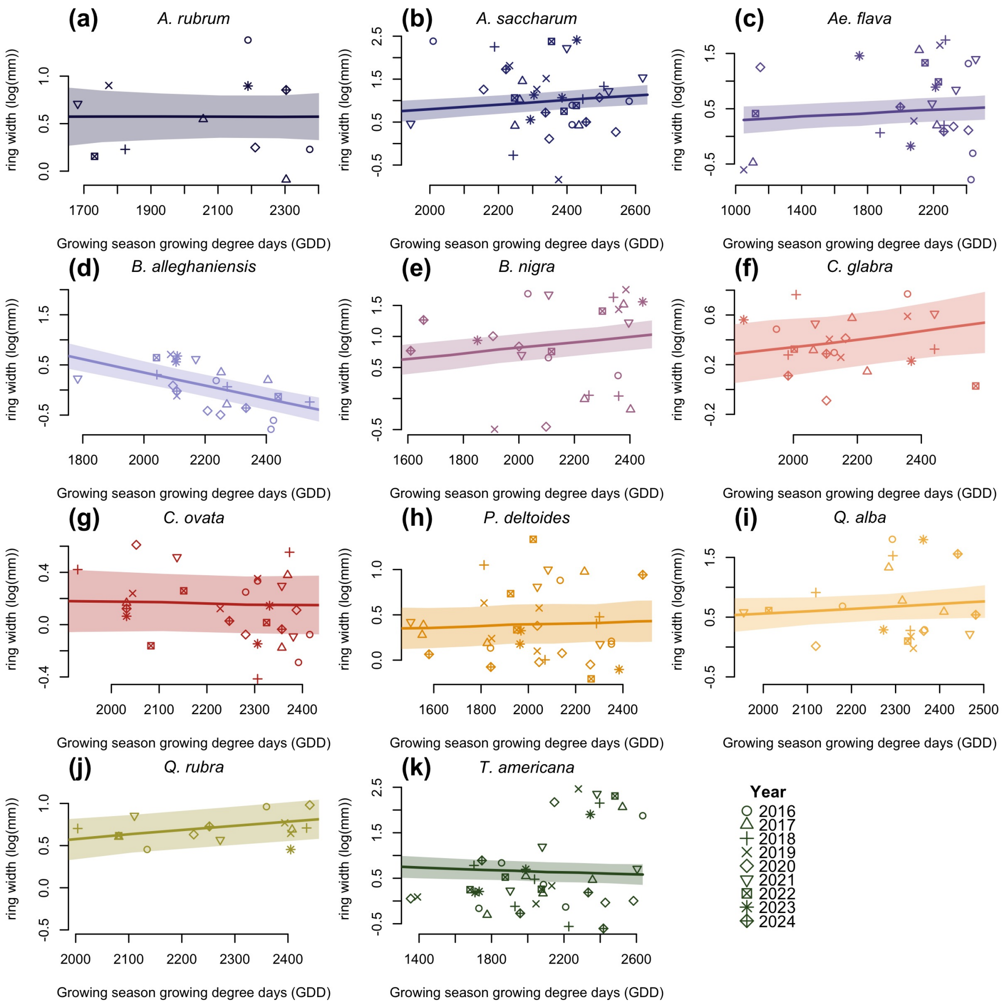

Figure S6: Ring width trends (log(mm)) in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

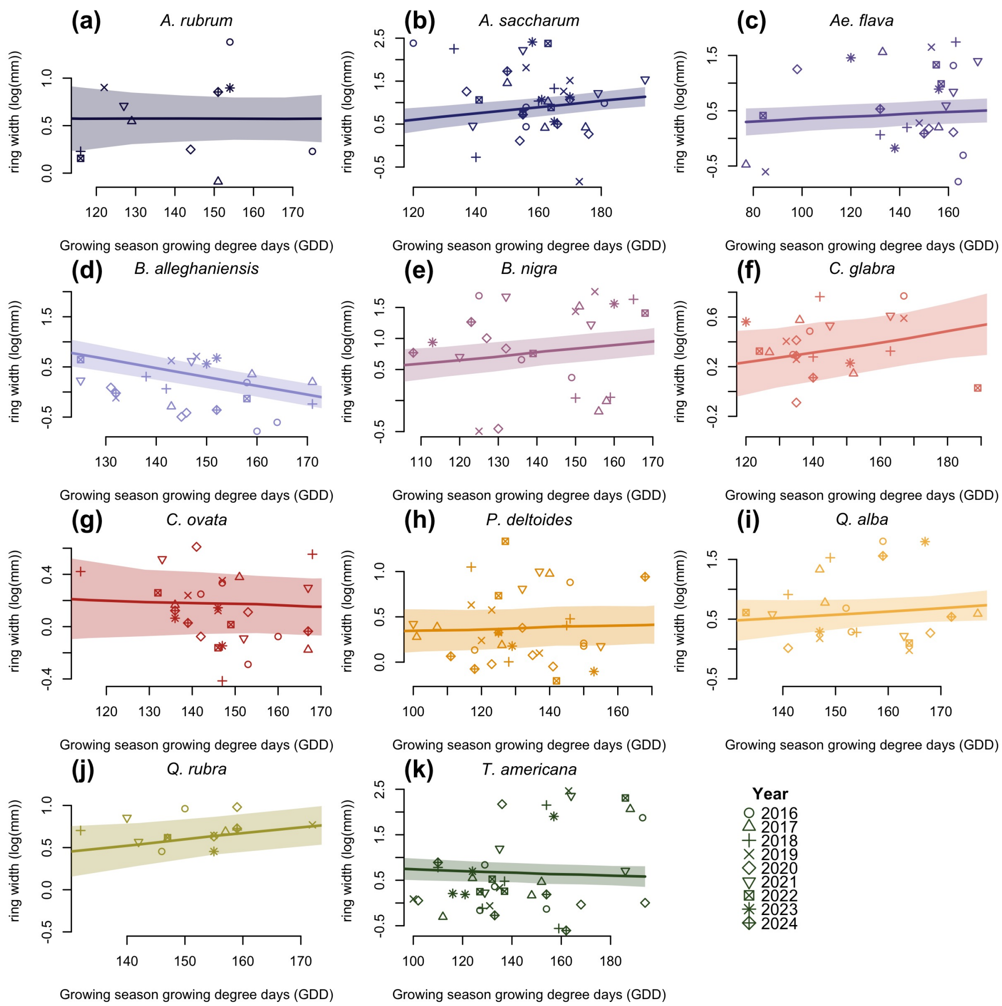

Figure S7: Ring width trends (log(mm)) in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

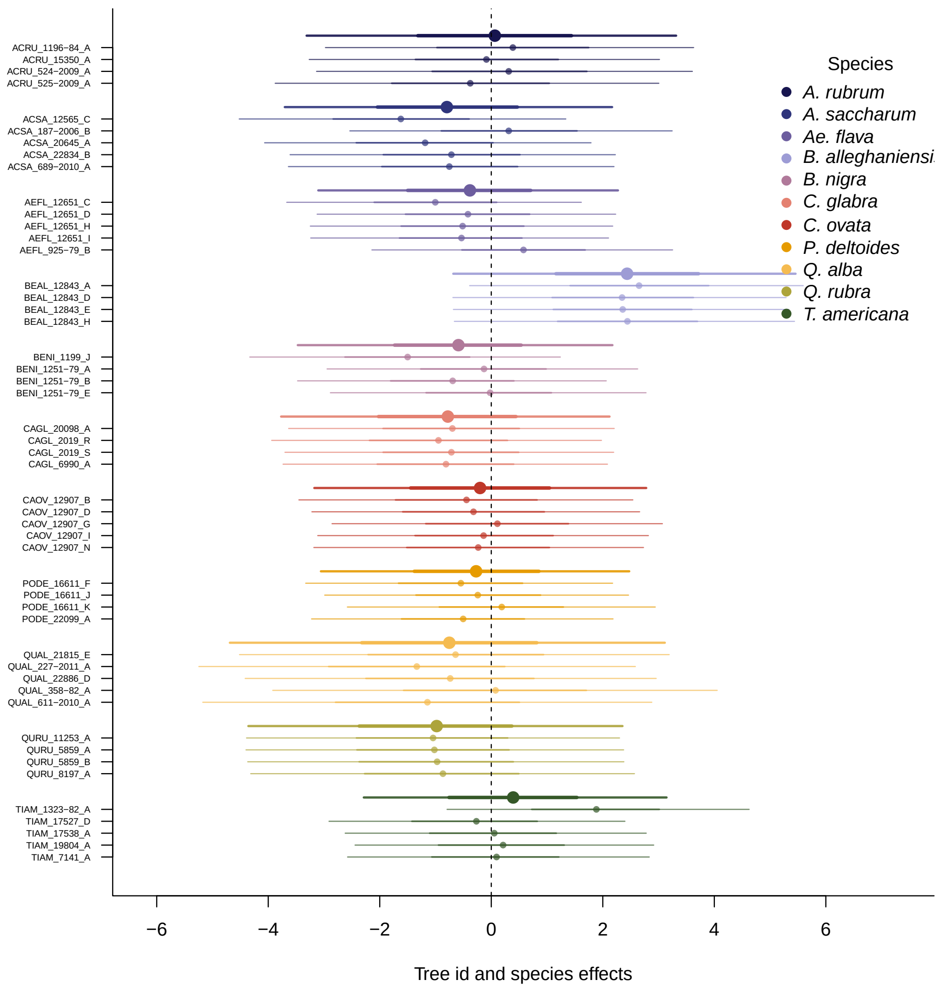

Figure S8: Ring width (log(mm)) deviations from the grand mean for each grouping factor of the GDD model. The different tree id values represent the combined intercepts of tree id, species and averaged year intercepts. The larger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

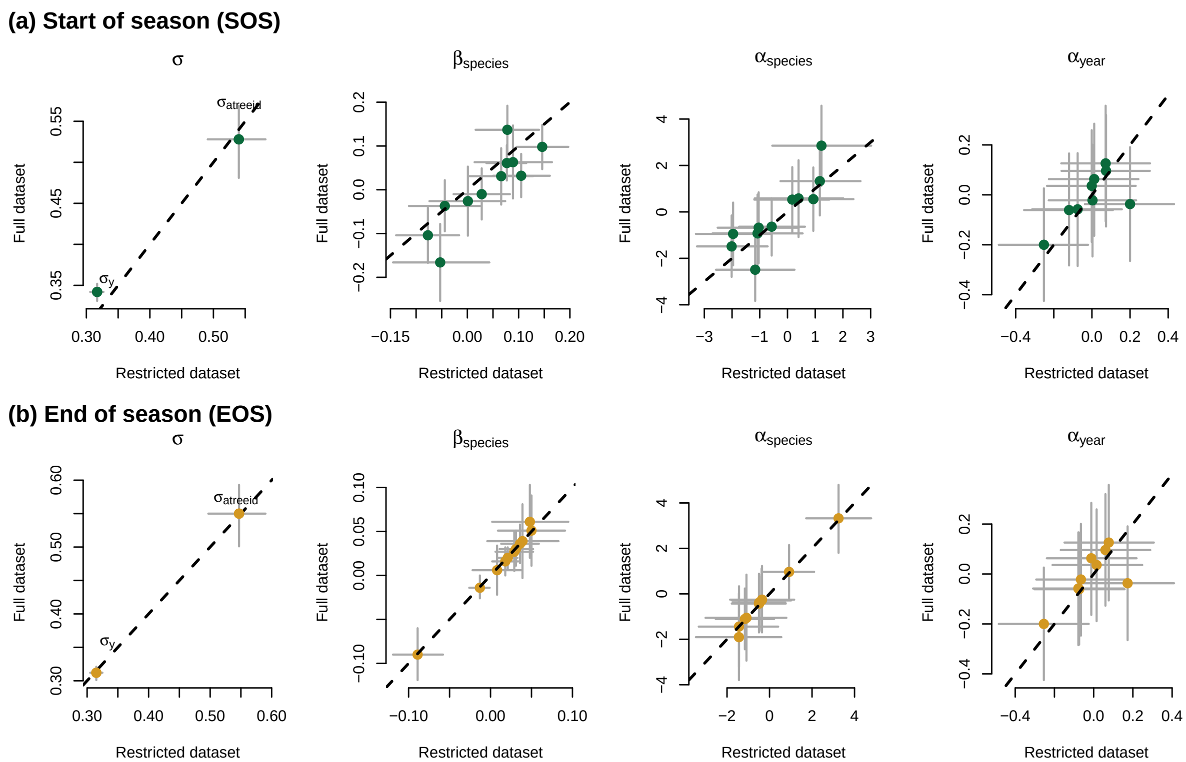

Figure S9: Full vs, most restricted rows of data for the model with start of season (SOS; a) and end of season (EOS; b) as the predictor. Each panel represent the different parameters used to fit the models. $\alpha_{species}$ and α_{year} represent the ring width (log(mm)) for species and year, respectively. $\beta_{species}$ represent the change in ring width (log(mm)) per scaled predictor for each species (see 'Data analyses' section in Methods for more details on predictor scaling).

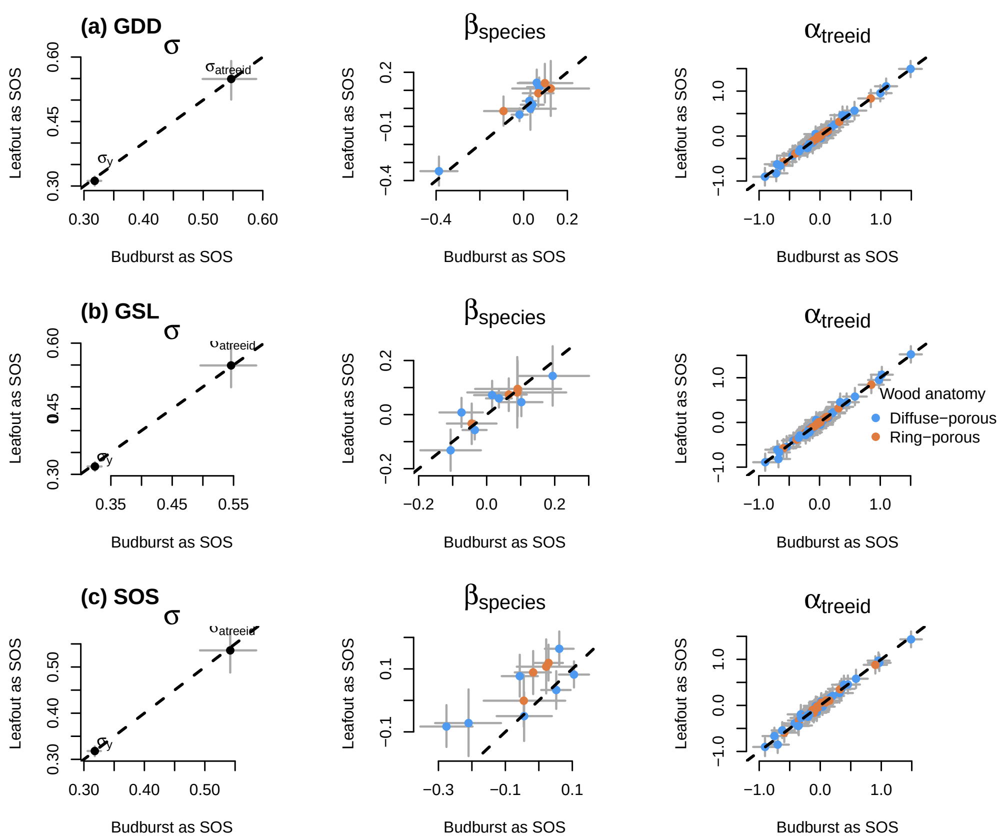

Figure S10: Parameter estimates when calculating the GDD (a), GSL (b) from leafout vs budburst, to leaf coloring. (c) shows the model estimates when taking leafout vs budburst as the SOS (see 'Data analyses' section in Methods for more details).

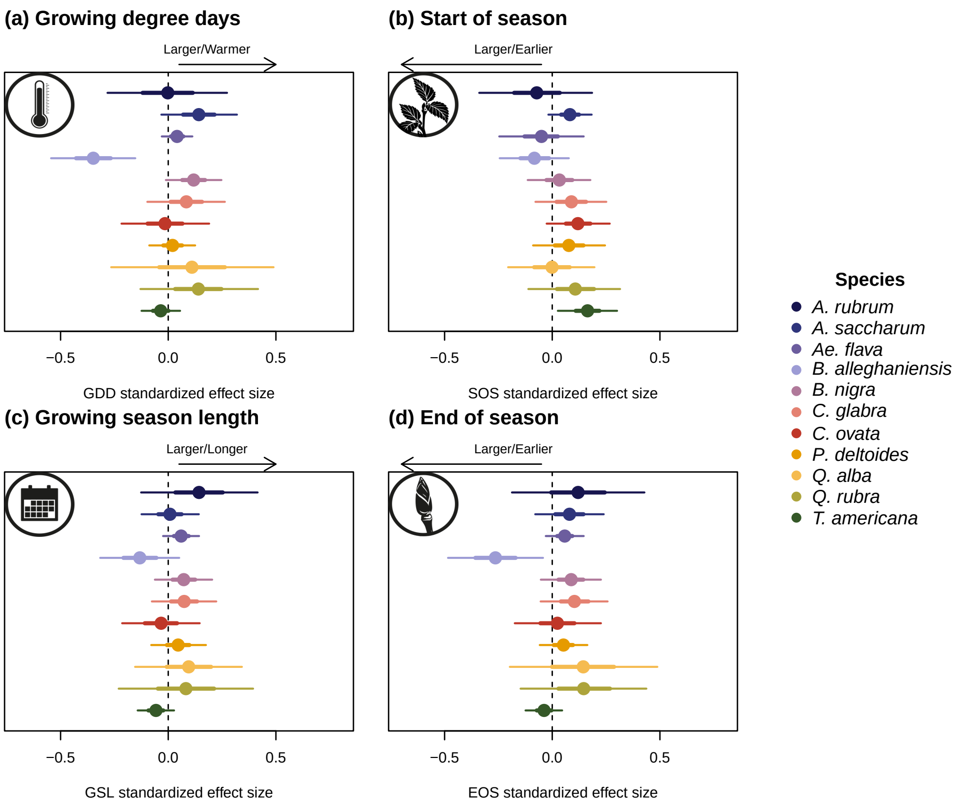

Figure S11: Effect size of each predictors across our eleven species. We depict ring width (log(mm)) change per scaled (z-scored; see 'Data analyses' section in Methods for more details) predictor for each species. We report estimates as mean and 90% uncertainty intervals (in square brackets) across our four different predictors (GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season).

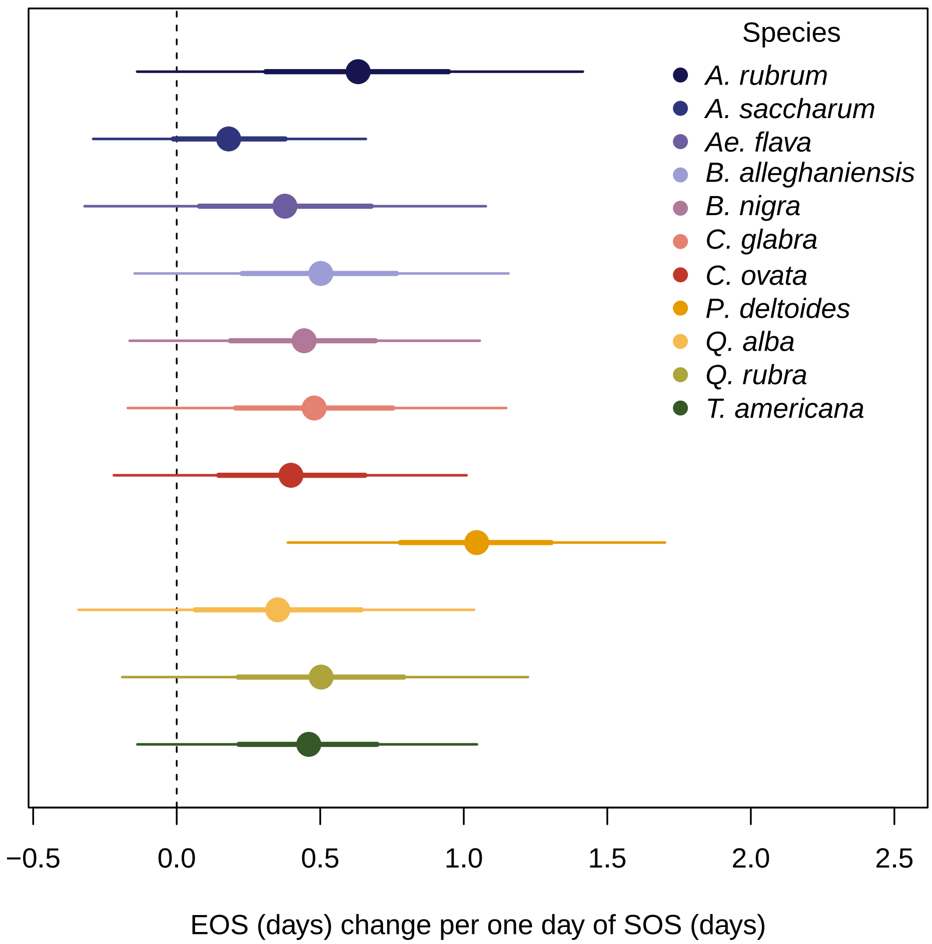

Figure S12: Effect of the start of season (SOS) on the end of season (EOS) across our eleven species. We report these slope estimates as mean and 90% uncertainty intervals.

Chapter 3

Concluding chapter

1. We used observational data to understand how longer and warmer seasons change growth for juvenile and mature trees
   (a) Our findings are significant as they show that forest recruitment may not be harmed by climate change
   (b) Mature trees store most of the forests carbon

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Appendix A

Supporting Materials

This would be any supporting material not central to the dissertation. For example:

- additional details of methodology and/or data;
- diagrams of specialized equipment developed.;
- copies of questionnaires and survey instruments.